SCIENTIFIC COMMITTEE
TWENTY-THIRD REGULAR SESSION

10–18 August 2027 (provisional)
Pohnpei, Federated States of Micronesia (provisional)

South Pacific albacore 2024: KIMSA assessment

WCPFC-SC23-2027/SA-WP-XX (DRAFT)
9 September 2026

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This demonstration rebuilds the South Pacific albacore 2024 assessment with KIMSA. It combines one diagnostic fit, eight cumulative full-workflow model-development fits, 100 independently refitted ensemble members, 30 starting-value trials, 20 simulation-estimation refits, five retrospective peels, five one-year index hindcasts, an average-adult-biomass likelihood profile and 5,000 reference-catch projection paths. All supplied finite results are carried into the displays; recorded numerical checks remain available in the interactive outputs.

The fitted period ends in 2022 and projections cover 2023–2042. Management summaries use 2019–2022 for recent spawning quantities and 2018–2021 for recent fishing mortality.

The supplementary interactive viewer provides the detailed study, fitted-data and portable-result views from one entry page.

This report demonstrates a complete KIMSA assessment workflow for South Pacific albacore using the public 2024 assessment inputs. The analysis spans the stock from the equator to 50°S and retains the four-region assessment structure shown in Figure 1. The fitted time series ends in 2022.

The diagnostic fit, model-development sequence, Monte Carlo ensemble, robustness studies and projections are connected through portable KIMSA result contracts. The public Pacific Community repositories were used as read-only sources; this demonstration does not modify or replace the official assessment.

The 2024 assessment builds on the spatial structure used for the preceding South Pacific albacore assessment. WCPO regions are separated at 10°S and 25°S, and the eastern Pacific is represented as a fourth region. The model-development and ensemble inputs follow the 2024 assessment and its published ensemble archive (Pacific Community, 2024; Teears et al., 2024).

The purpose of this workflow is to keep each fitted model, diagnostic study, figure, table and report linked to its exact input and package revision. Model outputs are reported as supplied; the interactive viewer exposes the underlying series and fits.

Catch, abundance indices, length compositions, weight compositions and conditional age-at-length observations are carried from the assessment inputs into the diagnostic result. Table 1 records the fishery roles, regions, observation periods and units used by the model.

Abundance-index panels pair each observation with its fitted value, conditional uncertainty band and residual. Composition panels retain the original samples and their model predictions. Records are grouped only for display; the fitted model uses the complete observations provided to its doitall workflow.

The diagnostic model uses the four-region, age- and size-structured ALB 2024 inputs with the `mfcl_no_tag` KIMSA specification. Catch is conditioned through the hybrid-F solver with five passes and a maximum F of 6. Each independent model starts from its declared input and executes the complete phase 1–9 doitall sequence.

0.1 KIMSA assessment software

KIMSA (Kit for Integrated Modular Stock Assessment) is the RTMB-based engine used to assemble and fit the population, fishing and observation components in this analysis. RTMB supplies automatic differentiation and Laplace-approximation tools to the fitted model (Kristensen, 2026). `kimsa.lab` runs the stepwise, ensemble and diagnostic studies; `kimsa.view` turns their validated result tables into publication figures and one interactive viewer; and `kimsa.report` assembles those outputs into this editable manuscript. The report reads portable result contracts, so rendering does not require an MFCL executable or a live fit.

0.2 Stepwise development

Eight cumulative model configurations were each fitted through their complete declared doitall sequence. Colours identify configurations and all trajectories use solid lines. The four panels in Figure 6 show spawning depletion, aggregate fishing mortality, recruitment and spawning potential.

0.3 Model ensemble

The ensemble contains 100 member-specific official 2024 inputs. Each member was refitted independently with KIMSA before its derived quantities were combined.

Thirty phase-aware starting-value trials perturb eligible optimizer coordinates when they first become active and then finish the unchanged doitall. Their objective, gradient and trajectory results retain every finite refit and identify recorded check status separately.

Twenty simulation-estimation tests simulate the declared observation components from the diagnostic KIMSA fit and refit each generated data set through the full workflow. Figure 13 summarises relative recovery error for derived quantities and available key parameter blocks, while Figure 14 shows trajectory spread.

Five retrospective peels, five one-year-ahead index hindcasts and a two-arm likelihood profile examine time-series stability, prediction and objective support. In the hindcast panels, the dashed series is predicted using only data before each held-out year; closer tracking and points nearer the one-to-one line indicate better out-of-sample prediction. Bias near zero and smaller RMSE are preferred; MASE below one is reported only when a valid training-history naive scale is available. The profile starts at the diagnostic centre; lower and upper arms advance sequentially in 2.5% steps within each arm and run concurrently with each other.

0.4 Data fit

Observed abundance indices, fitted values, supplied log-scale uncertainty and standardised residuals are paired by index in Figure 7. Composition figures use the observation and prediction records carried by the diagnostic fit.

0.5 Population estimates

The diagnostic population trajectories are shown in Figure 20. Regional shares are calculated only where regional quantities sum to the supplied all-region value; they are not inferred from fishery labels.

0.6 Stock status

Terminal ensemble uncertainty is shown in Kobe and Majuro coordinates in Figure 27. Annual status in Figure 28 compares the diagnostic path with the year-specific median across all ensemble members. Individual member paths remain in the supplementary viewer. Management quantities and reference-point probabilities use the explicit periods stated in the summary and are reported in Table 2 and Table 3.

The reference-catch projection follows the 2024 assessment specification and spans 2023–2042. Each of 100 independently refitted ensemble members contributes 50 stochastic recruitment paths. Figure 30 joins compatible historical ensemble summaries to the projected period, marks the transition, shows 2.5%–97.5% intervals and overlays three coherent individual paths selected across the terminal depletion distribution. Regional and catch-attainment results remain in the portable result tables. Terminal quantities are listed in Table 5.

The diagnostic, stepwise and ensemble views answer different questions. The diagnostic shows one detailed fit, the cumulative stepwise sequence isolates reproducible model choices, and the independently refitted ensemble describes variation among the supplied 2024 model members. Median trajectories and central run ranges summarise the ensemble without hiding the member-level paths retained in the viewer.

Starting-value, simulation-estimation, retrospective, hindcast and likelihood-profile results should be read together. They expose optimisation sensitivity, recovery under the fitted model, time-series dependence, predictive behaviour and local objective shape. The projections are a 20-year workflow demonstration under the declared reference-catch scenario and carry model-member and stochastic recruitment variation forward.

The public assessment design and write-up were provided by the Pacific Community. All model fitting, diagnostic assembly, visualisation and report generation in this demonstration use the pinned KIMSA companion package revisions recorded with the outputs.

1 References

Kristensen, K. (2026). RTMB: R bindings for TMB. https://doi.org/10.32614/CRAN.package.RTMB.

Pacific Community (2024). ALB 2024 ensemble results. GitHub repository https://github.com/PacificCommunity/ofp-sam-alb-2024-ensemble.

Teears, T., Castillo-Jordán, C., Davies, N., Day, J., Hampton, J., Magnusson, A., Peatman, T., Pilling, G., Xu, H., Vidal, T., Williams, P., and Hamer, P. (2024). Stock assessment of south pacific albacore: 2024. WCPFC-SC20-2024/SA-WP-02, Rev. 3, Western; Central Pacific Fisheries Commission Scientific Committee https://meetings.wcpfc.int/node/23119.

2 Tables

Table 1: Fishery definitions and observation units.

Fishery	Fishery name	Region	Role	Years	Unit
1	LL.DWFN.1a	1	Catch	1954–2022	fish
2	LL.DWFN.1b	1	Catch	1954–2022	fish
3	LL.DWFN.1c	1	Catch	1954–2022	fish
4	LL.DWFN.1d	1	Catch	1954–2022	fish
5	LL.PICT.1ab	1	Catch	1992–2022	fish
6	LL.PICT.1cd	1	Catch	1982–2022	fish
7	LL.AZ.1abcd	1	Catch	1987–2022	fish
8	LL.DWFN.1ef	1	Catch	1954–2022	fish
9	LL.PICT.1ef	1	Catch	1984–2022	fish
10	LL.AZ.1ef	1	Catch	1985–2022	fish
11	TR.ALL.1e	1	Catch	1982–2022	t
12	TR.ALL.1f	1	Catch	1987–2022	t
13	DN.ALL.1ef	1	Catch	1983–1990	t
14	LL.EPO.2a	2	Catch	1957–2022	fish
15	LL.EPO.2b	2	Catch	1961–2021	fish
16	LL.EPO.2c	2	Catch	1963–2021	fish
17	TR.EPO.2abc	2	Catch	1986–2006	t
18	LL.INDEX.1abcd.1954-2022	1	Index	1954–2022	relative
19	TR.INDEX.1ef.NZ-troll	1	Index	1992–2022	relative
20	LL.INDEX.2abc.1954-2022	2	Index	1954–2022	relative

Table 2: Management quantities and supplied uncertainty summaries.

Quantity	Period	Unit	Mean	Median	Minimum	10%	90%	Maximum
Aggregate F	2018–2021	per year	0.0728	0.0666	0.0222	0.0332	0.108	0.156
Aggregate F	2022	per year	0.0692	0.0619	0.0206	0.0314	0.104	0.151
BMSY	Reference point	t	785000	662000	330000	423000	1290000	2210000
Dynamic unfished spawning potential	2019–2022	t	678000	668000	601000	624000	748000	813000

Quantity	Period	Unit	Mean	Median	Minimum	10%	90%	Maximum
Dynamic unfished spawning potential	2022	t	705000	690000	616000	651000	799000	901000
F/F_{MSY}	2018–2021	ratio	0.458	0.448	0.205	0.253	0.638	0.85
F/F_{MSY}	2022	ratio	0.435	0.429	0.189	0.232	0.613	0.818
FMSY	Reference point	per year	0.155	0.159	0.106	0.123	0.182	0.203
MSY	Reference point	t/year	115000	103000	62900	75100	177000	235000
$SB/SB_{F=0}$	2019–2022	ratio	0.513	0.512	0.274	0.39	0.668	0.742
$SB/SB_{F=0}$	2022	ratio	0.548	0.542	0.292	0.422	0.703	0.781
SB/SB_{MSY}	2019–2022	ratio	3.22	3.03	1.55	2.1	4.89	6.01
SB/SB_{MSY}	2022	ratio	3.58	3.32	1.67	2.28	5.47	6.97
SBMSY	Reference point	t	114000	114000	60600	83600	146000	167000
Spawning potential	2019–2022	t	349000	335000	222000	262000	488000	604000
Spawning potential	2022	t	389000	369000	238000	287000	546000	704000

Table 3: Stock-status probabilities for the declared reference points.

Condition	Period	Probability
$F/F_{MSY} > 1$	2018–2021	0
$SB/SB_{F=0} < 0.20$	2019–2022	0
$SB/SB_{MSY} < 1$	2019–2022	0
$F/F_{MSY} > 1$	2022	0
$SB/SB_{F=0} < 0.20$	2022	0
$SB/SB_{MSY} < 1$	2022	0

Table 4: Model reference points and their definitions.

Quantity	Estimate	Unit
MSY	86055	t/year
FMSY	0.14846	per year
BMSY	579660	t
SBMSY	136940	t

Table 5: Projected quantities by scenario and reporting period.

Scenario	Period	Quantity	Median	95% interval
Reference catch	2042	Spawning Biomass	273700	7.053e+04–6.297e+05
Reference catch	2042	Dynamic Spawning Depletion	0.46049	0.1184–0.8391
Reference catch	2042	Fishing Mortality Proxy	0.084509	0.02722–0.2258
Reference catch	2042	Recruitment	90321000	1.728e+07–5.534e+08
Reference catch	2042	Total Biomass	943750	2.862e+05–2.997e+06
Reference catch	2042	Catch	79733	5.664e+04–8.432e+04

Table 6: Held-out prediction bias (predicted minus observed), RMSE and MASE by series and forecast lead. N counts scored forecasts; MASE uses each cutoff's training naive-error scale.

Series	Lead	N	Bias	RMSE	MASE	Unit
Longline index / Region 1	1	5	0.013635	0.075785	0.35625	index
New Zealand troll index / Region 1	1	5	0.43361	0.70298	2.4447	index
Longline index / Region 2	1	5	-0.13701	0.21829	0.84836	index

3 Figures

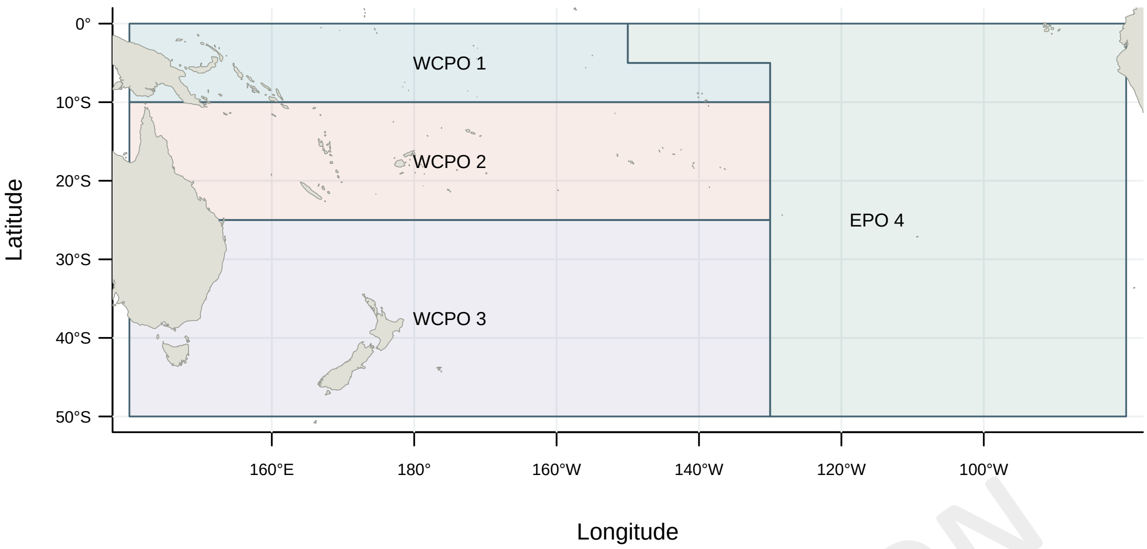

Figure 1: Four-region South Pacific assessment structure. Boundaries follow the published ALB assessment definition: 10°S and 25°S in the WCPFC Convention Area, with one eastern Pacific region.

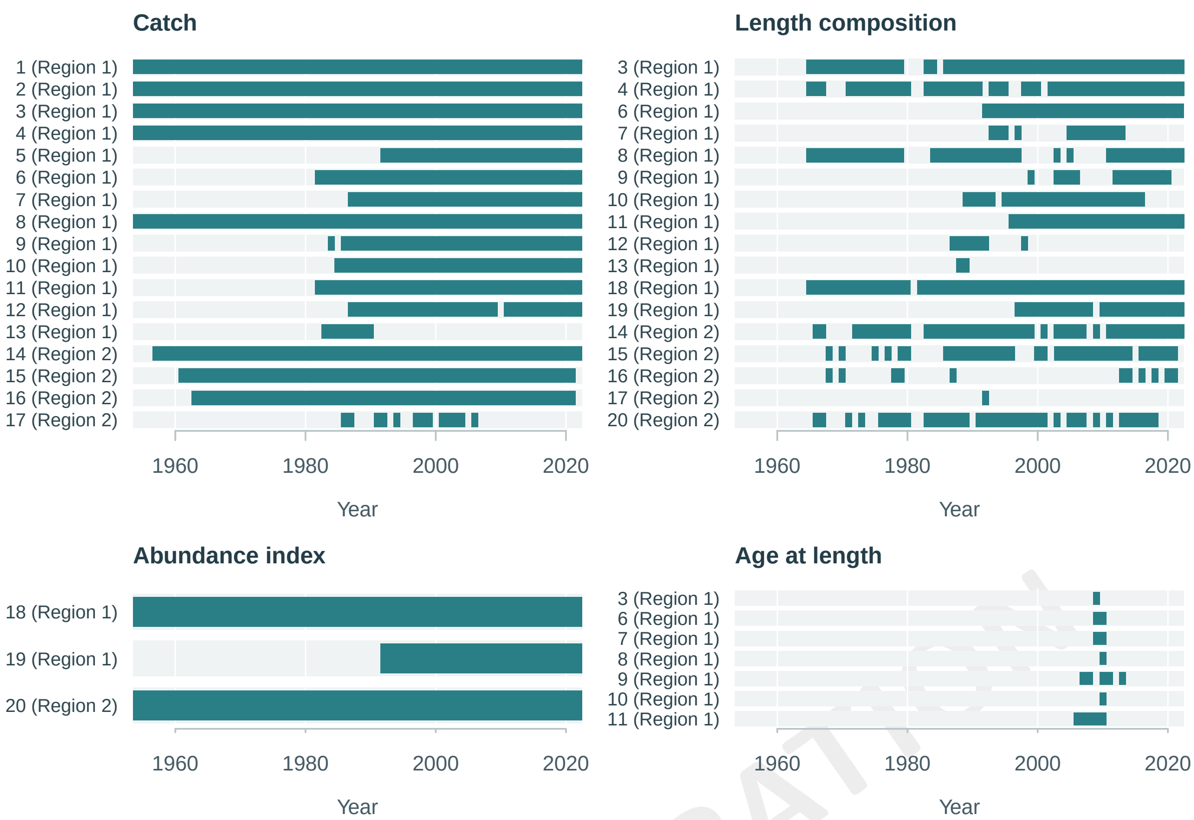

Figure 2: Observation availability by data type and fishery. Coloured cells indicate years with observations.

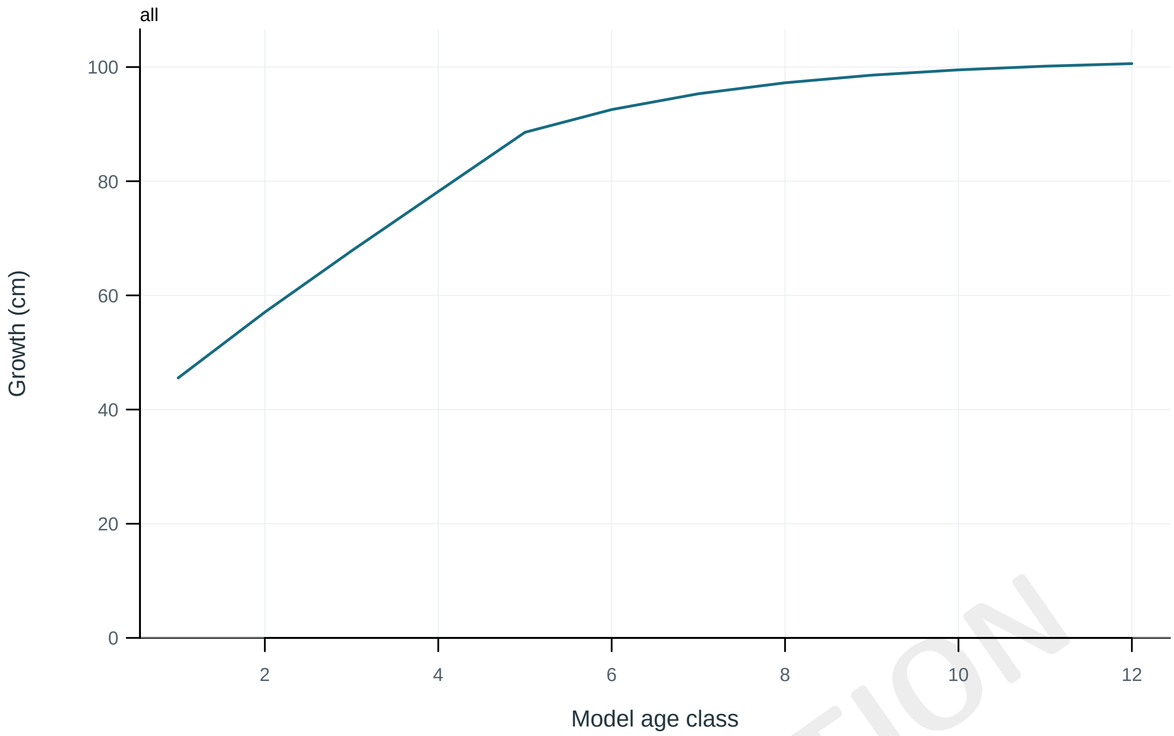

Figure 3: Mean length at the start of each model age class.

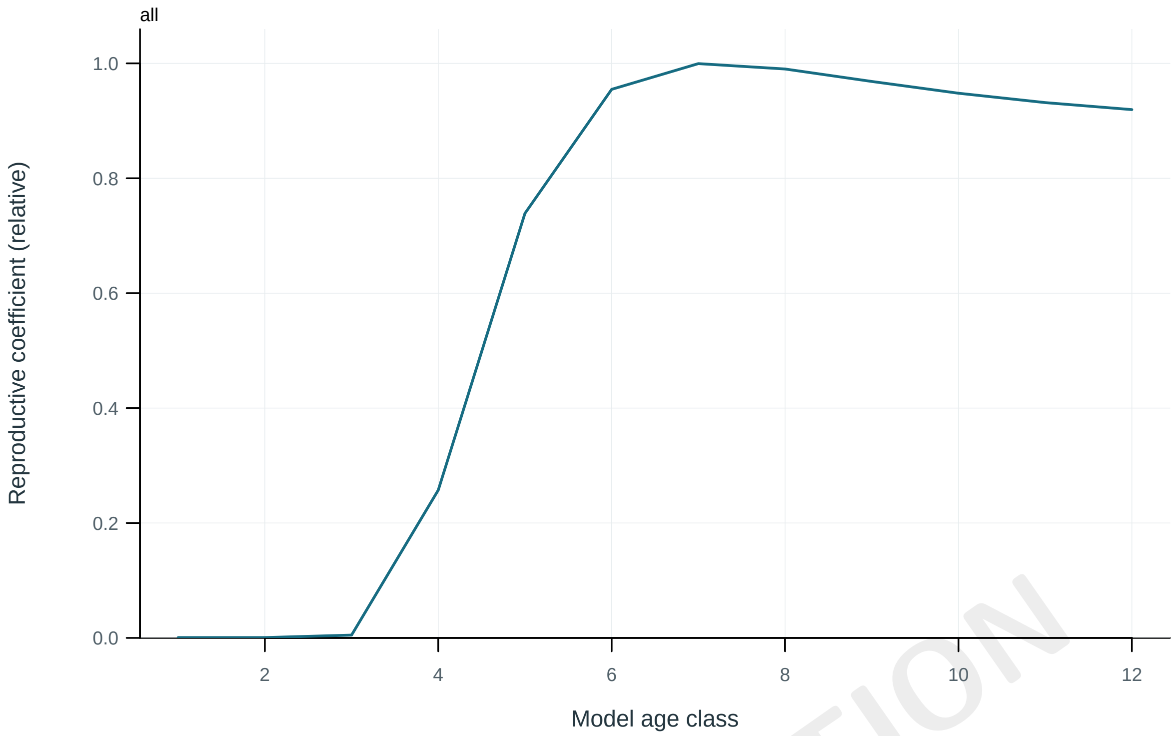

Figure 4: Model reproductive coefficient at age.

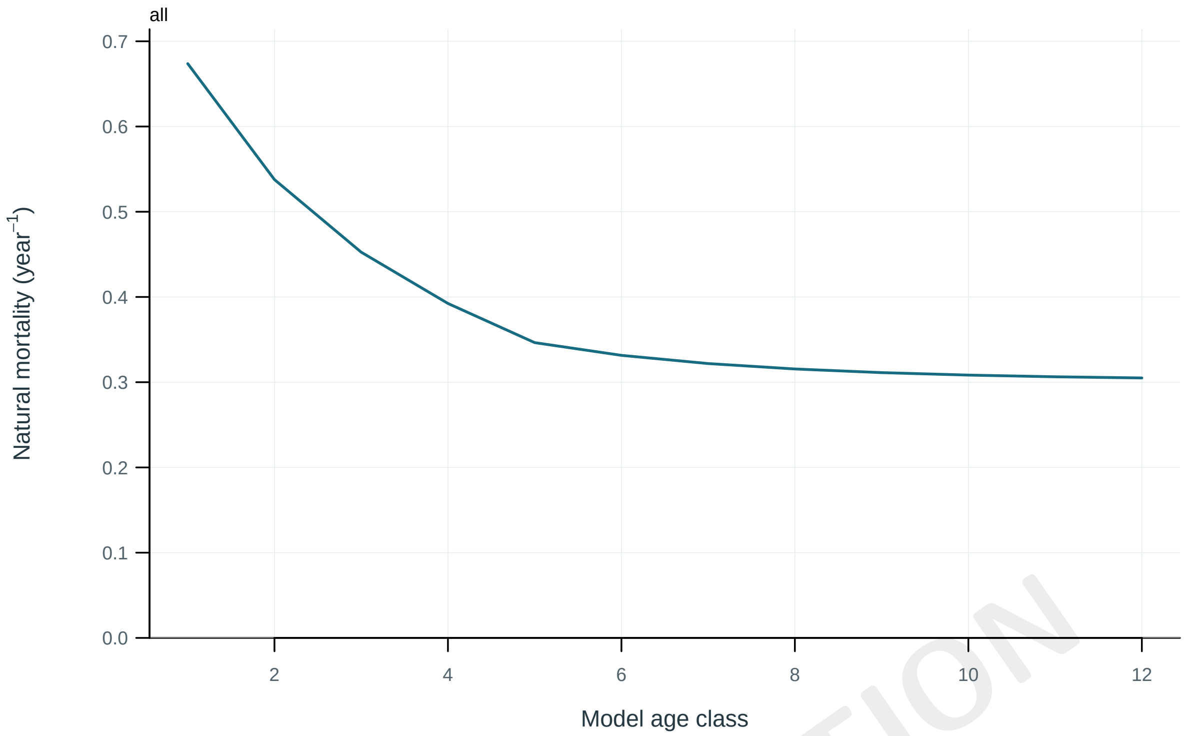

Figure 5: Annual instantaneous natural mortality at age.

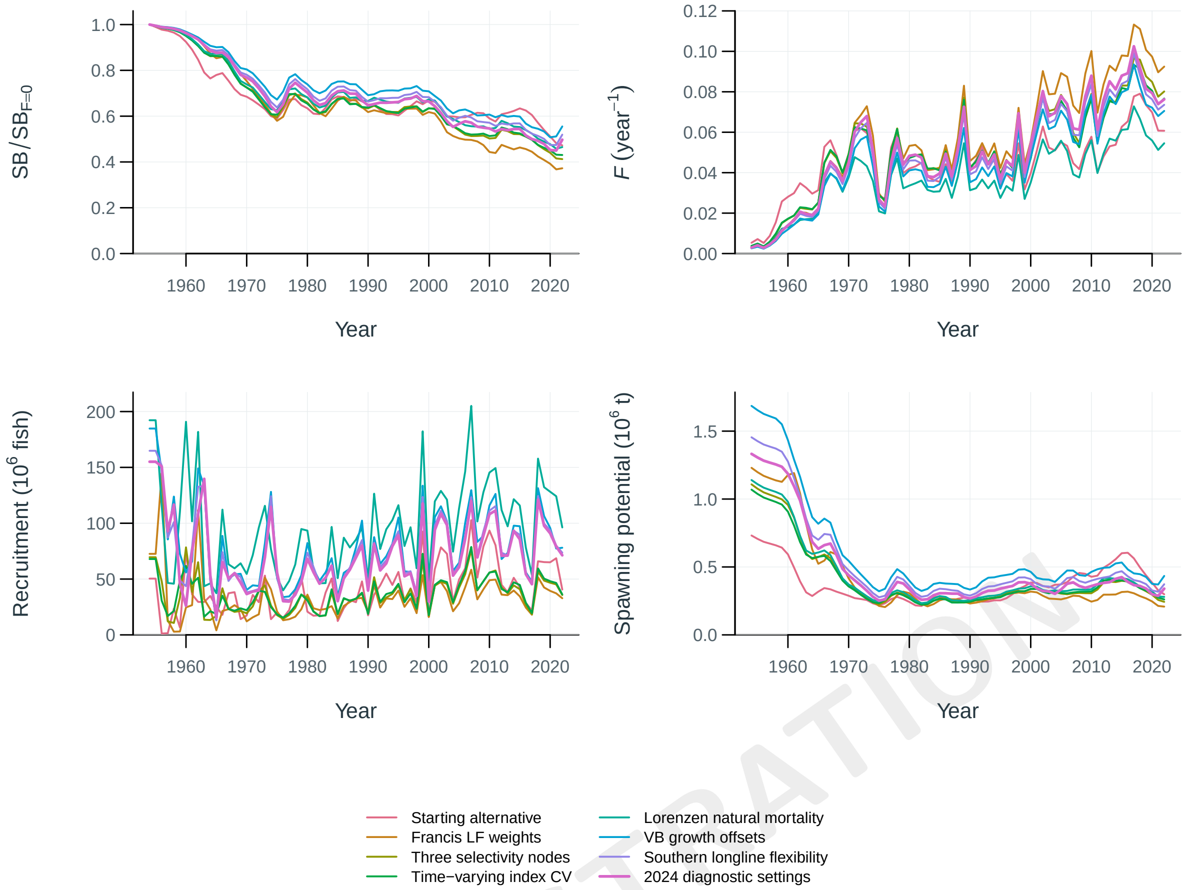

Figure 6: Stepwise model trajectories.

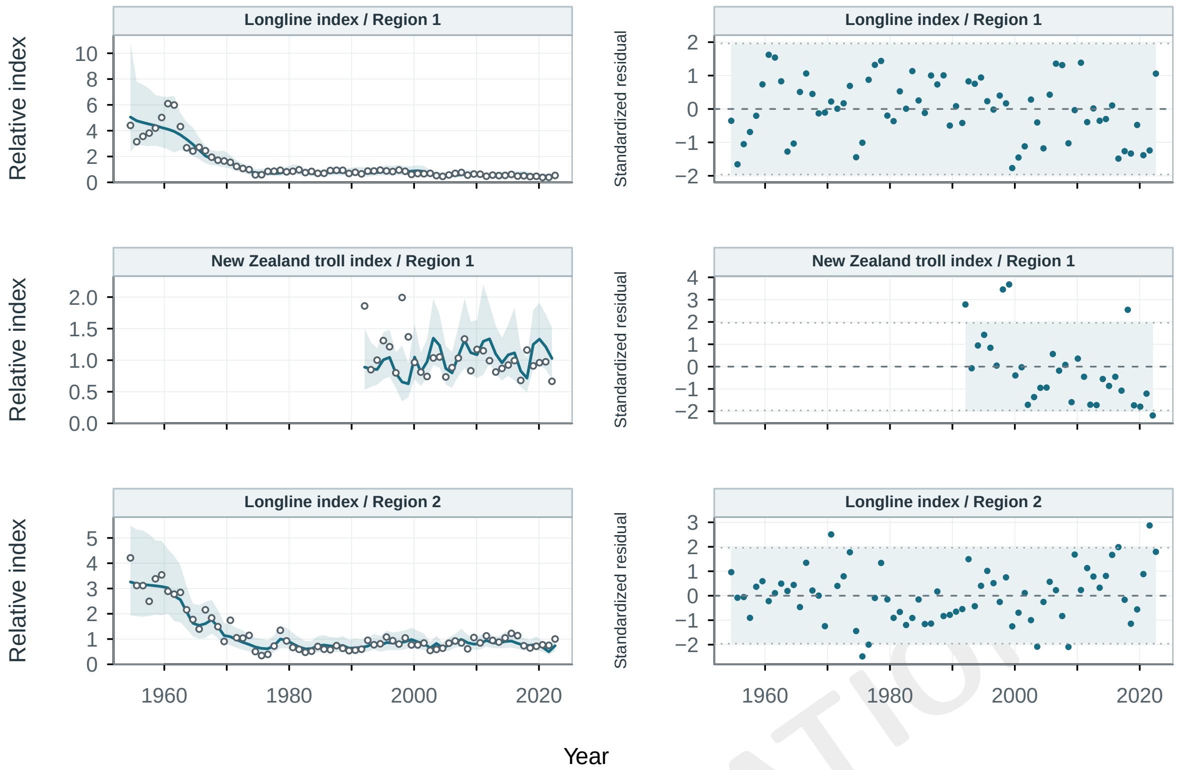

Figure 7: Abundance-index fits and log residuals by series. Shading: central 95% conditional observation intervals. Residuals use the supplied log-scale standard deviations.

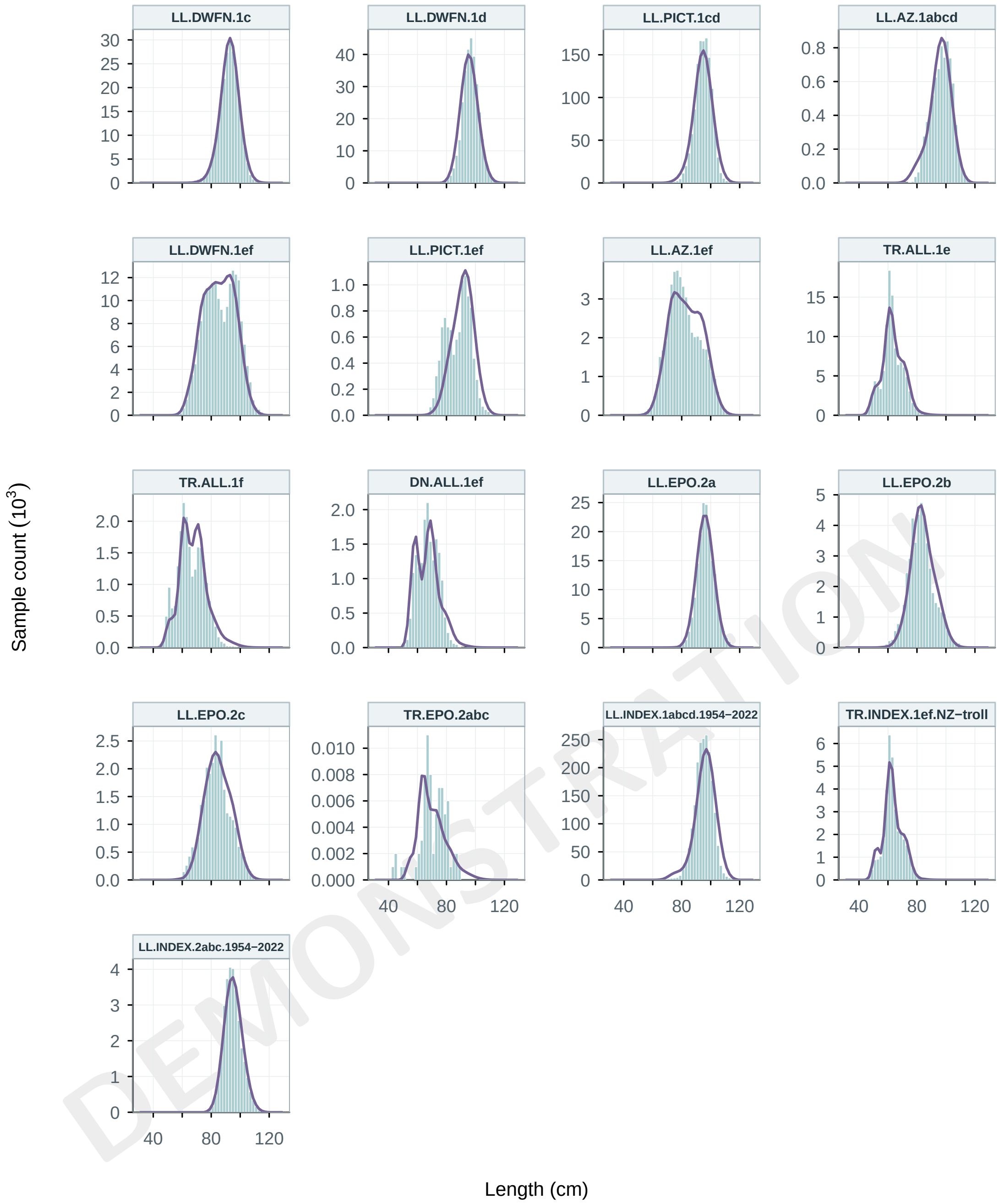

Figure 8: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

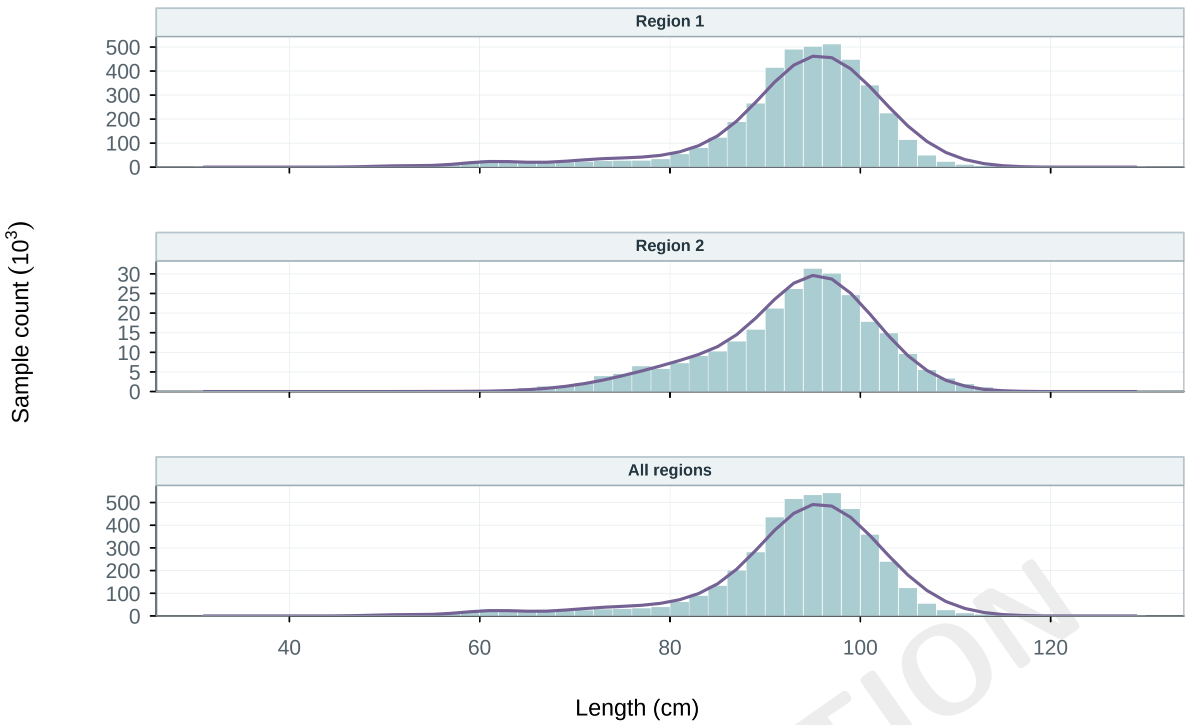

Figure 9: Length-composition fits by region. Observed sample counts (bars) and predictions (lines), pooled over years.

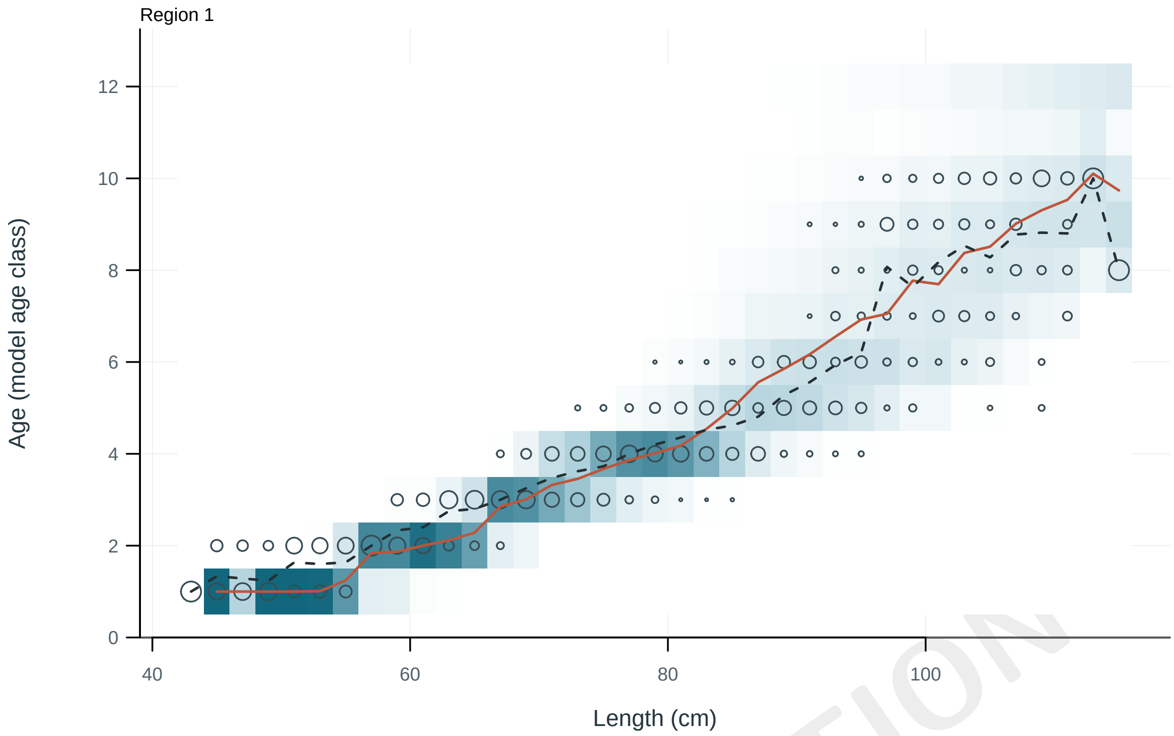

Figure 10: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

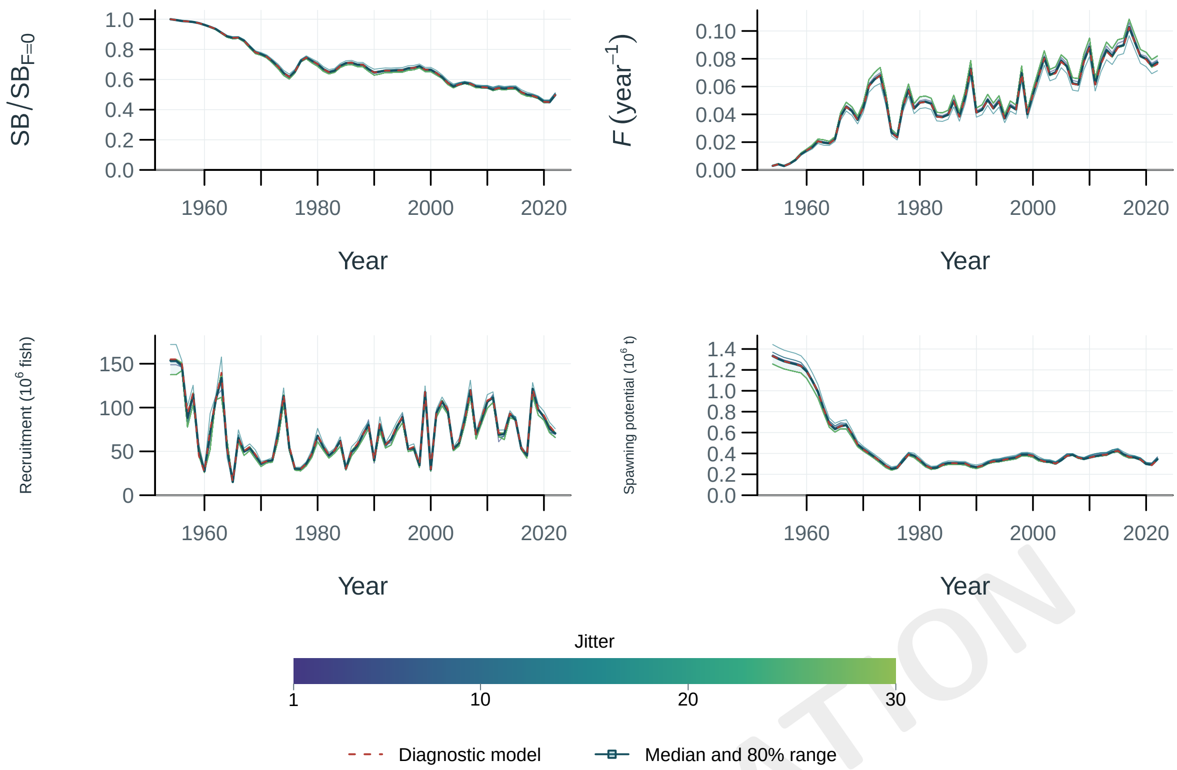

Figure 11: Population trajectories across models (thin lines) and the reference model (red). Shading shows central 80% pointwise run ranges; the dark line is the median and the dashed red line is the reference model. Where isolated outer ranges dominate, upward triangles mark values beyond the robust static axis; complete values remain in the viewer. The reference model is excluded from these summaries.

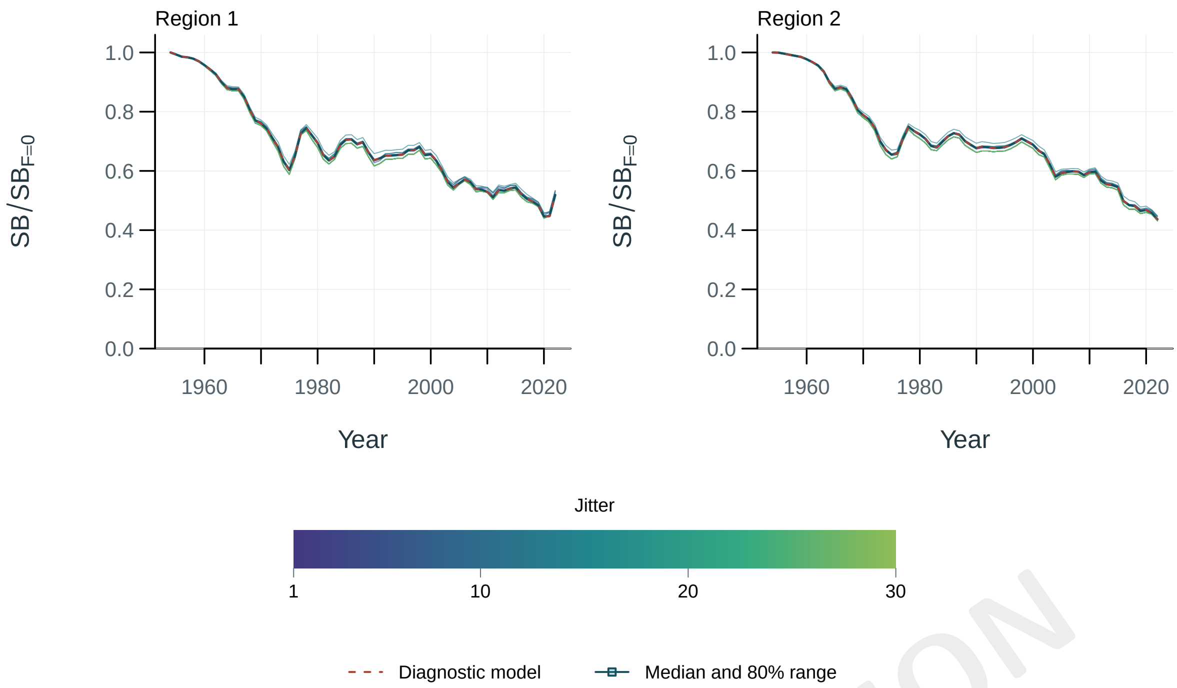

Figure 12: Spawning depletion by model and region, relative to each model's dynamic unfished reference. Shading shows central 80% pointwise run ranges; the dark line is the median and the dashed red line is the reference model. Where isolated outer ranges dominate, upward triangles mark values beyond the robust static axis; complete values remain in the viewer. The reference model is excluded from these summaries.

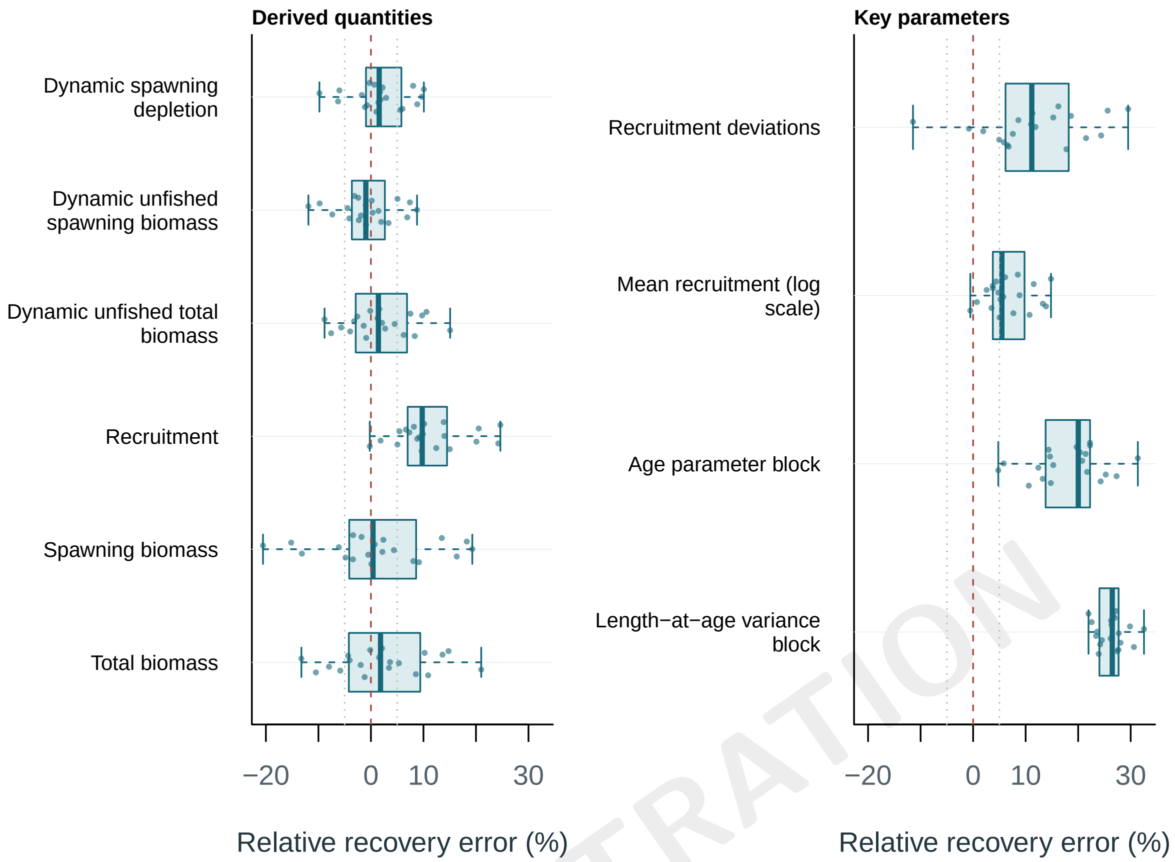

Figure 13: Relative recovery error for terminal quantities and supplied key parameters in simulated-data refits. Boxes summarize replicates and points retain every finite result.

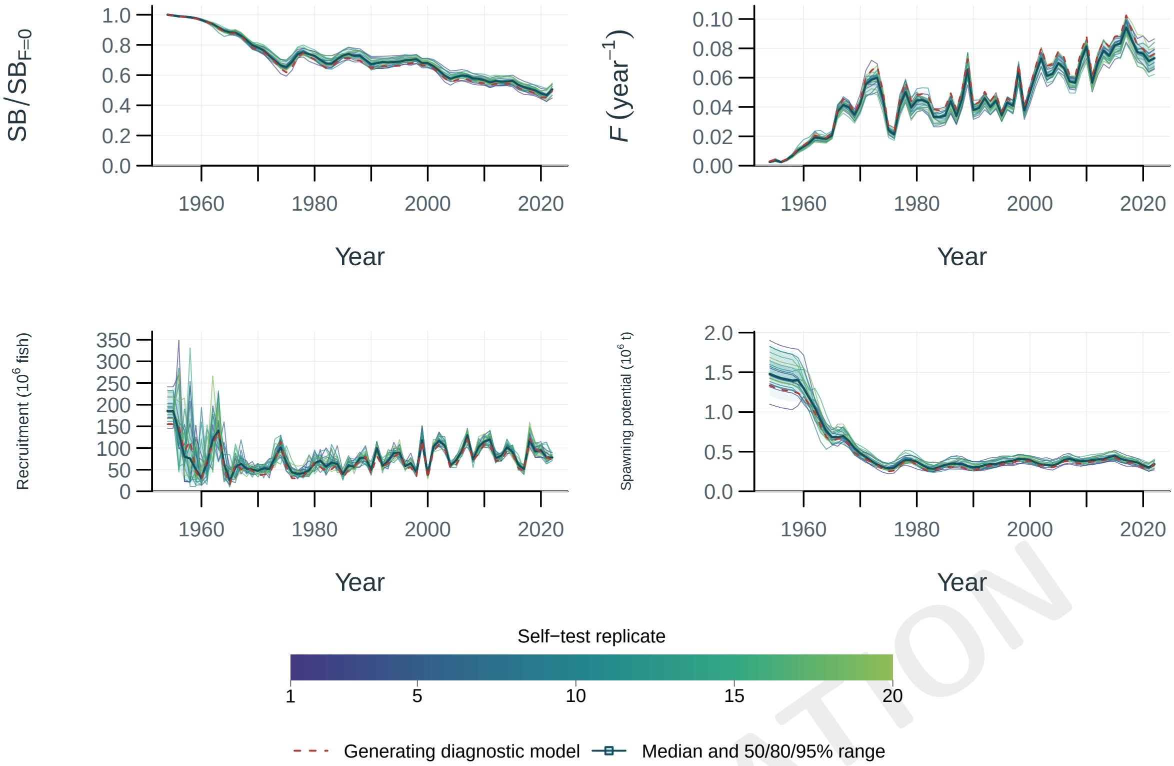

Figure 14: Population trajectories across models (thin lines) and the reference model (red). Shading shows central 50, 80, 95% pointwise run ranges; the dark line is the median and the dashed red line is the reference model. Where isolated outer ranges dominate, upward triangles mark values beyond the robust static axis; complete values remain in the viewer. The reference model is excluded from these summaries.

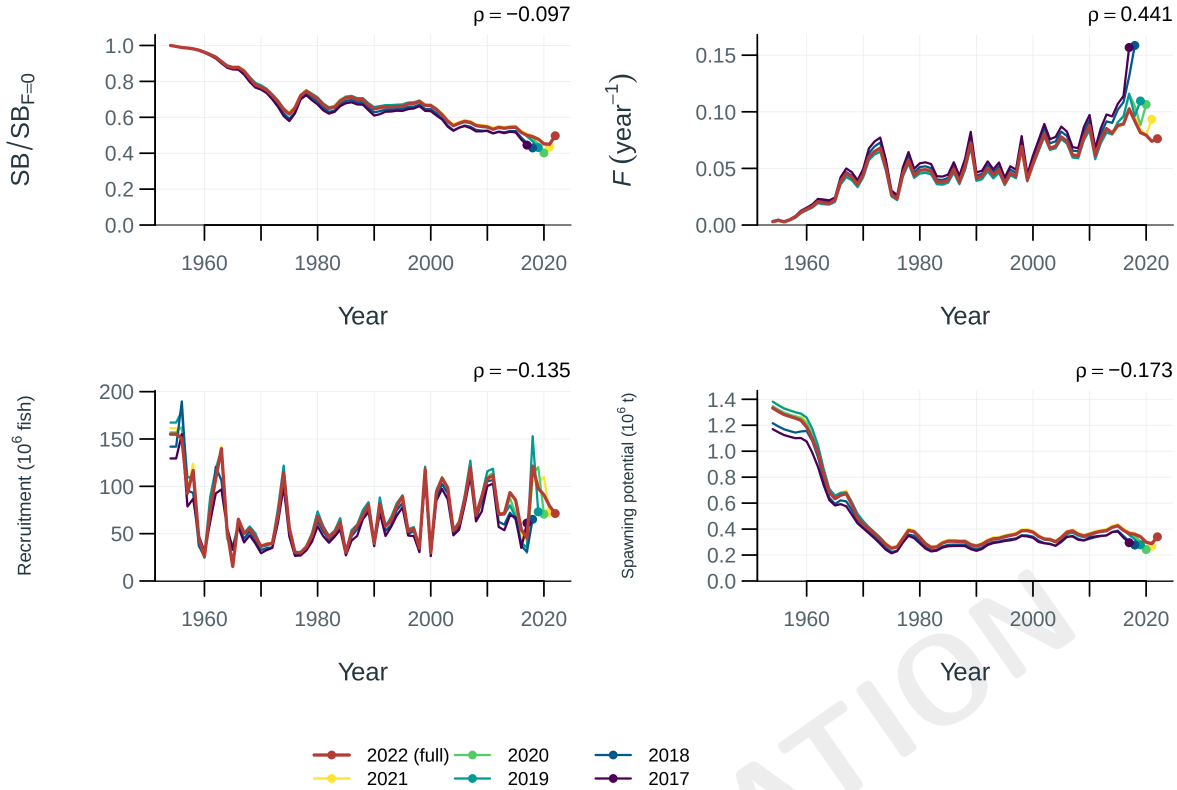

Figure 15: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

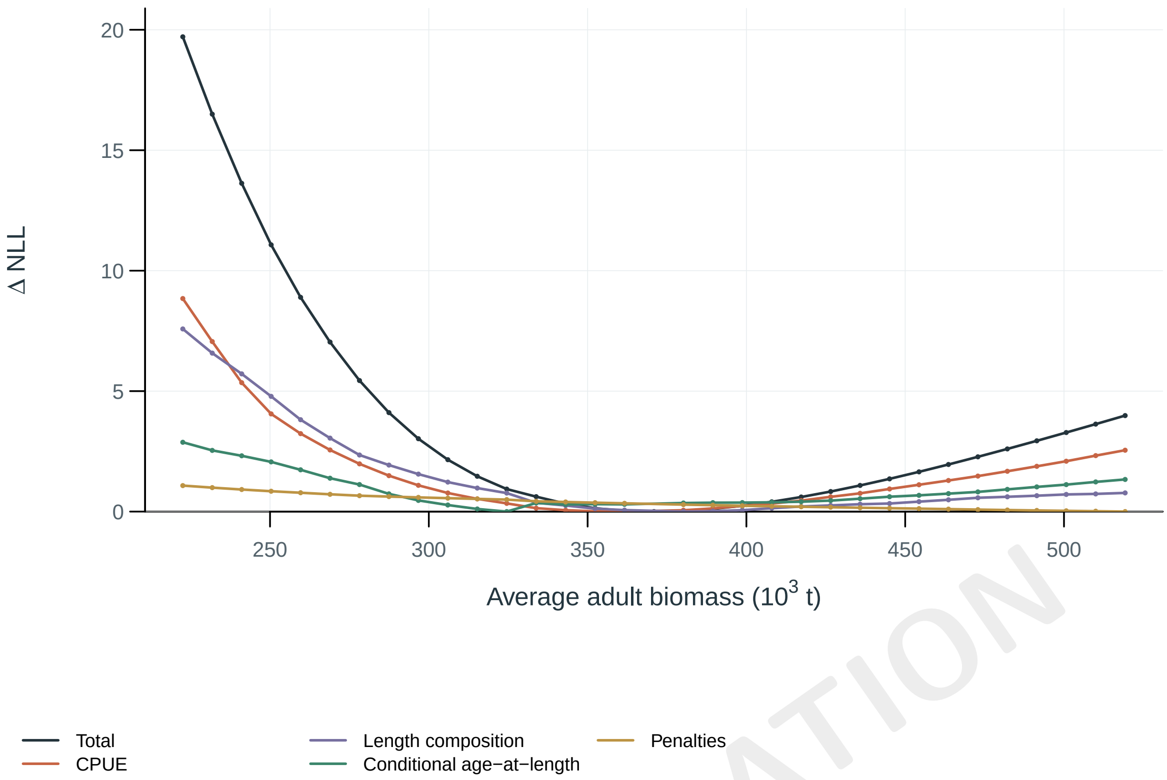

Figure 16: Total negative log-likelihood and component profiles. Each curve is centered on its own minimum. Reproductively weighted biomass summed over regions and averaged across model periods in 1954-2022.

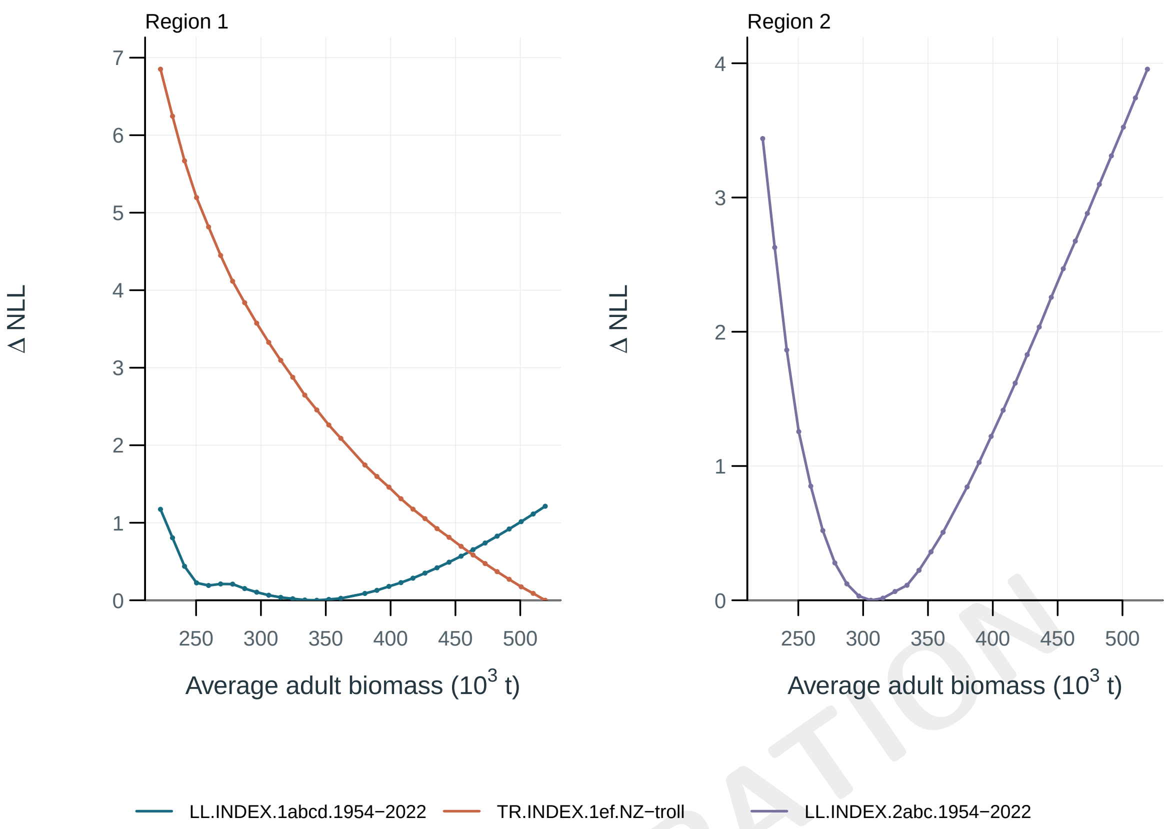

Figure 17: Negative log-likelihood profiles for abundance-index components. Each curve is centered on its own minimum. Reproductively weighted biomass summed over regions and averaged across model periods in 1954-2022.

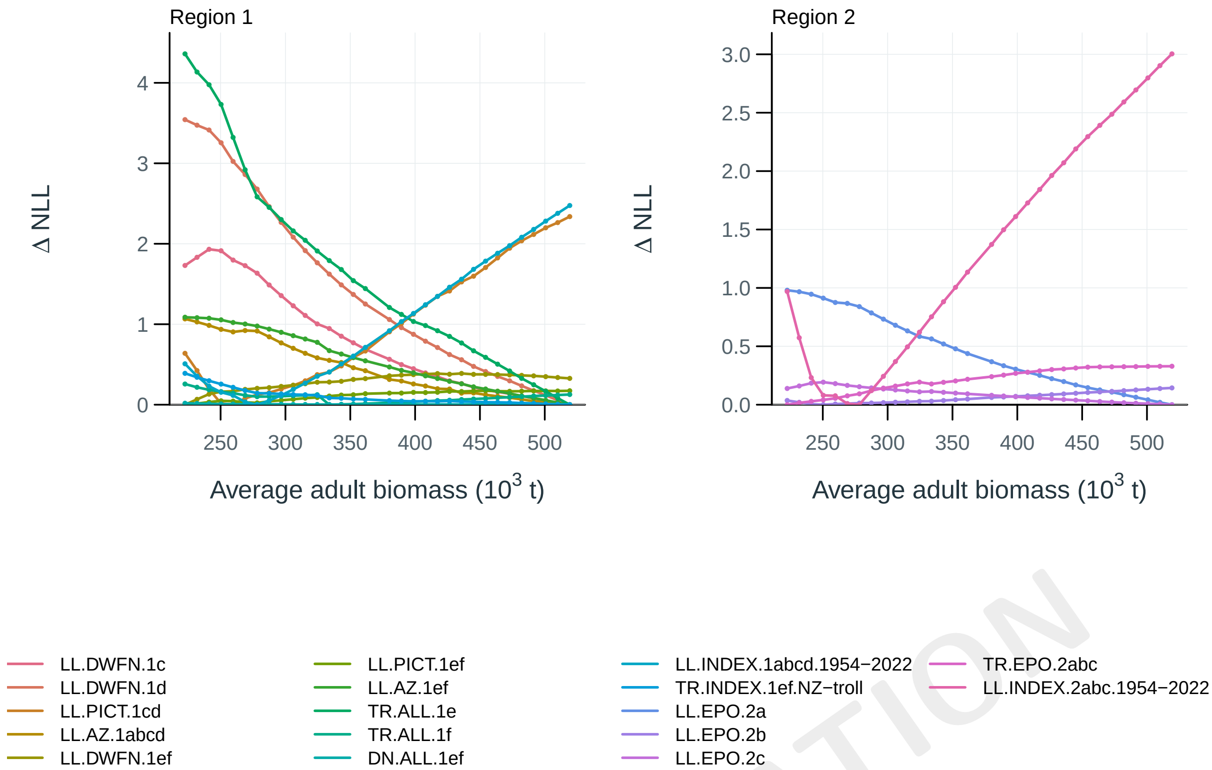

Figure 18: Negative log-likelihood profiles for length-composition components. Each curve is centered on its own minimum. Reproductively weighted biomass summed over regions and averaged across model periods in 1954-2022.

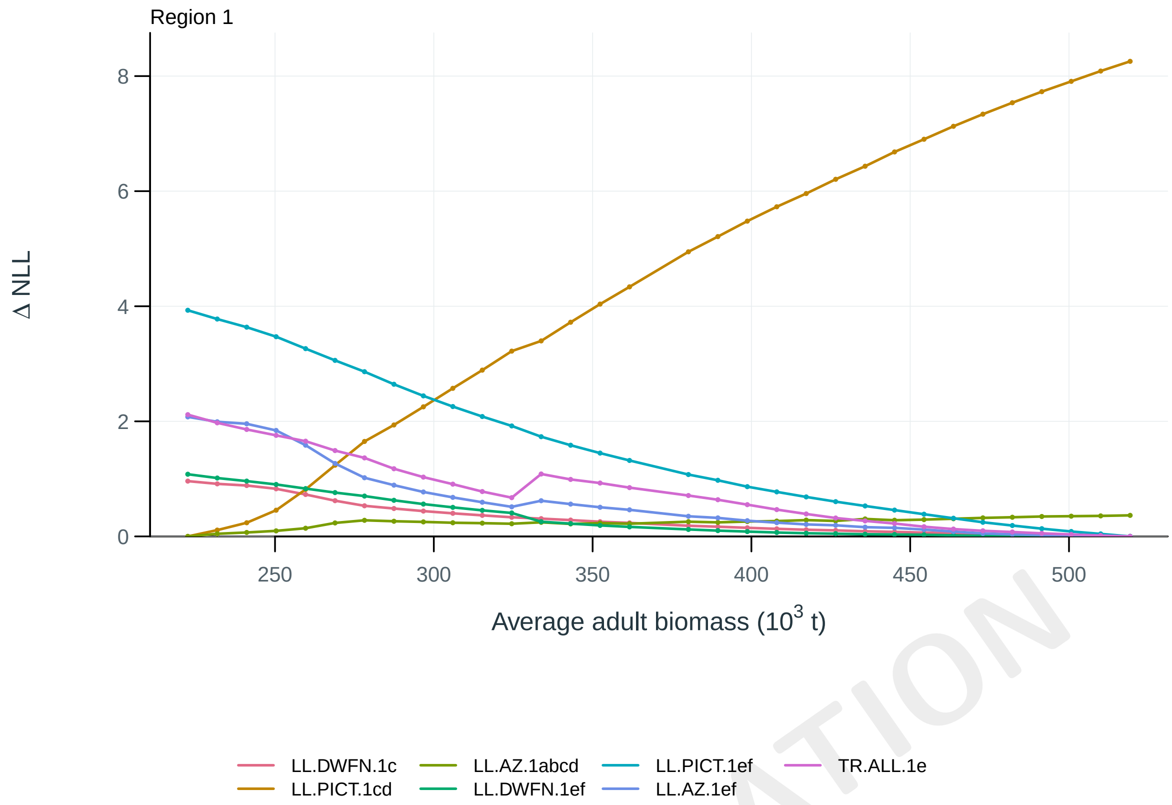

Figure 19: Negative log-likelihood profiles for conditional age-at-length components. Each curve is centered on its own minimum. Reproductively weighted biomass summed over regions and averaged across model periods in 1954-2022.

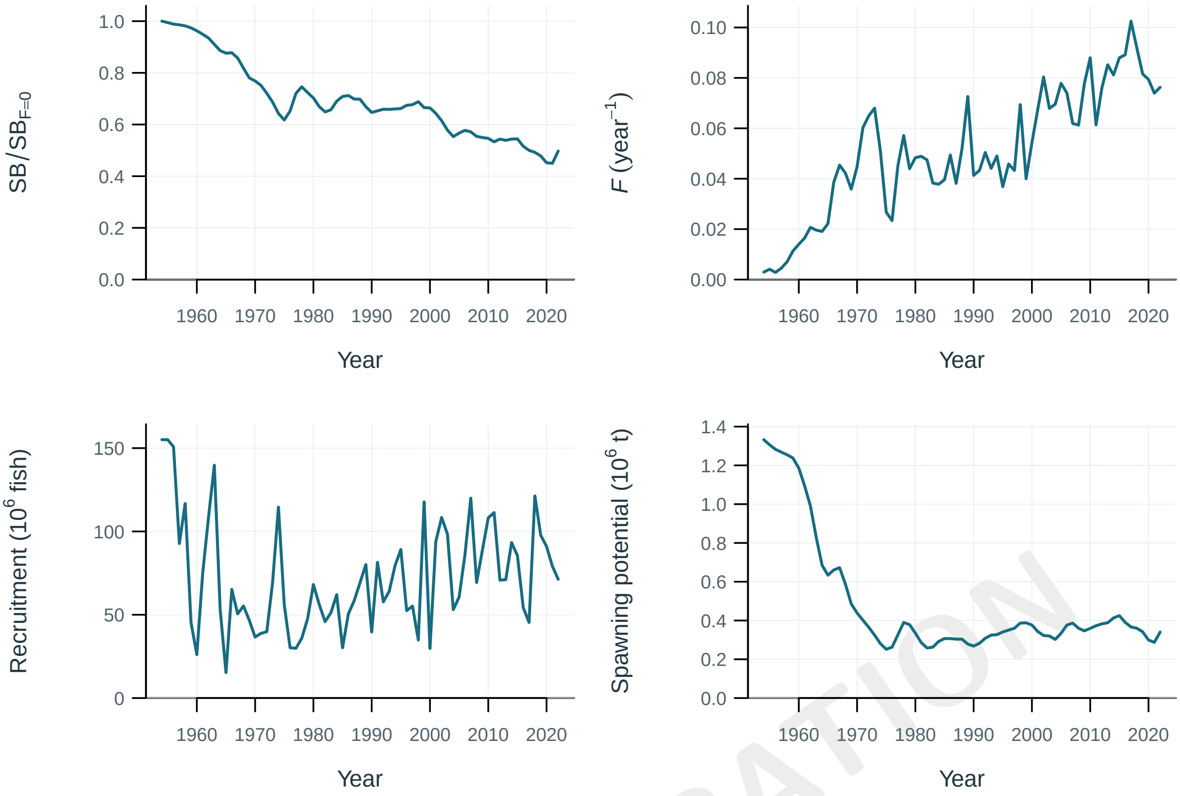

Figure 20: Estimated population trajectories.

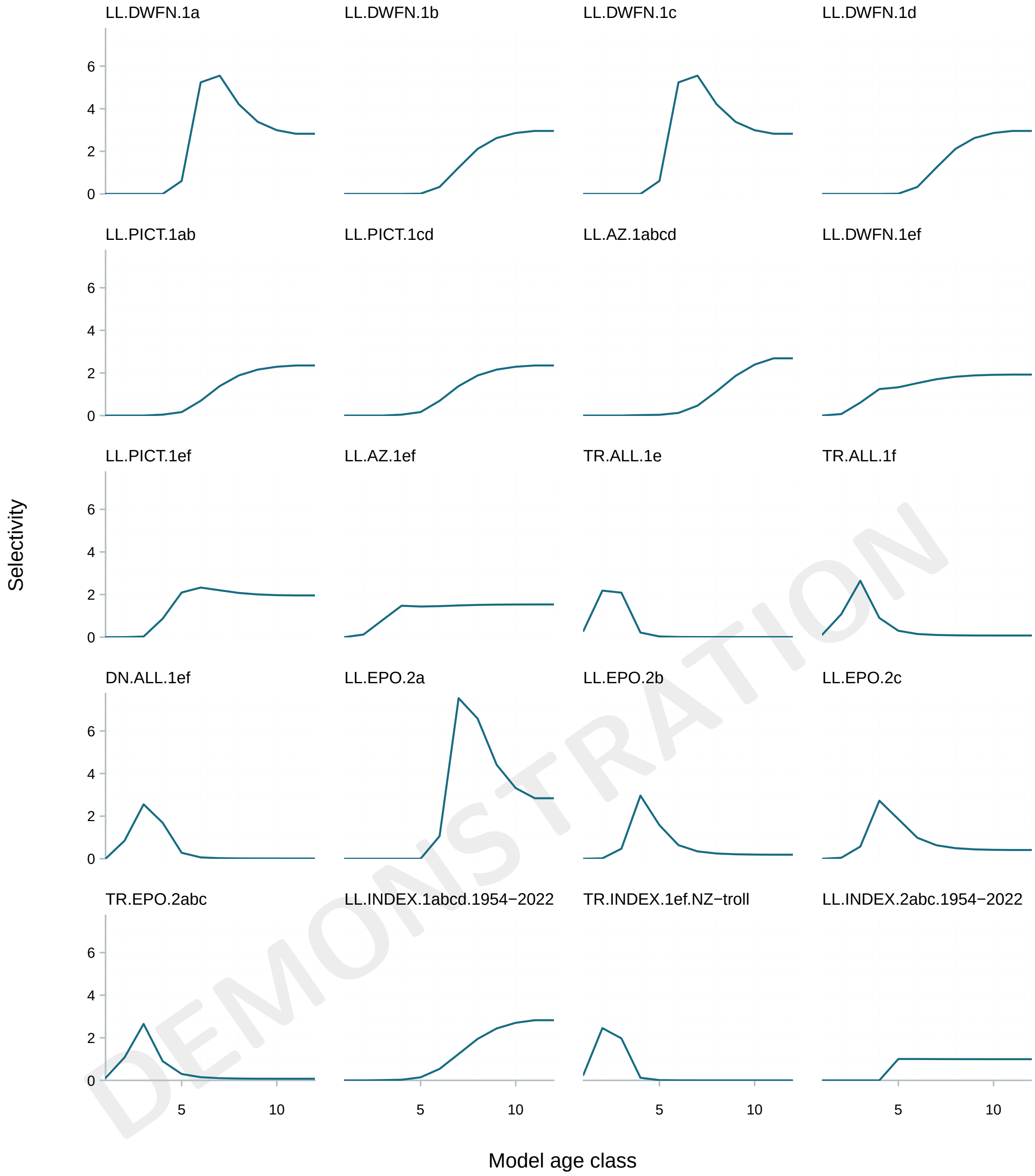

Figure 21: Fishery selectivity at age.

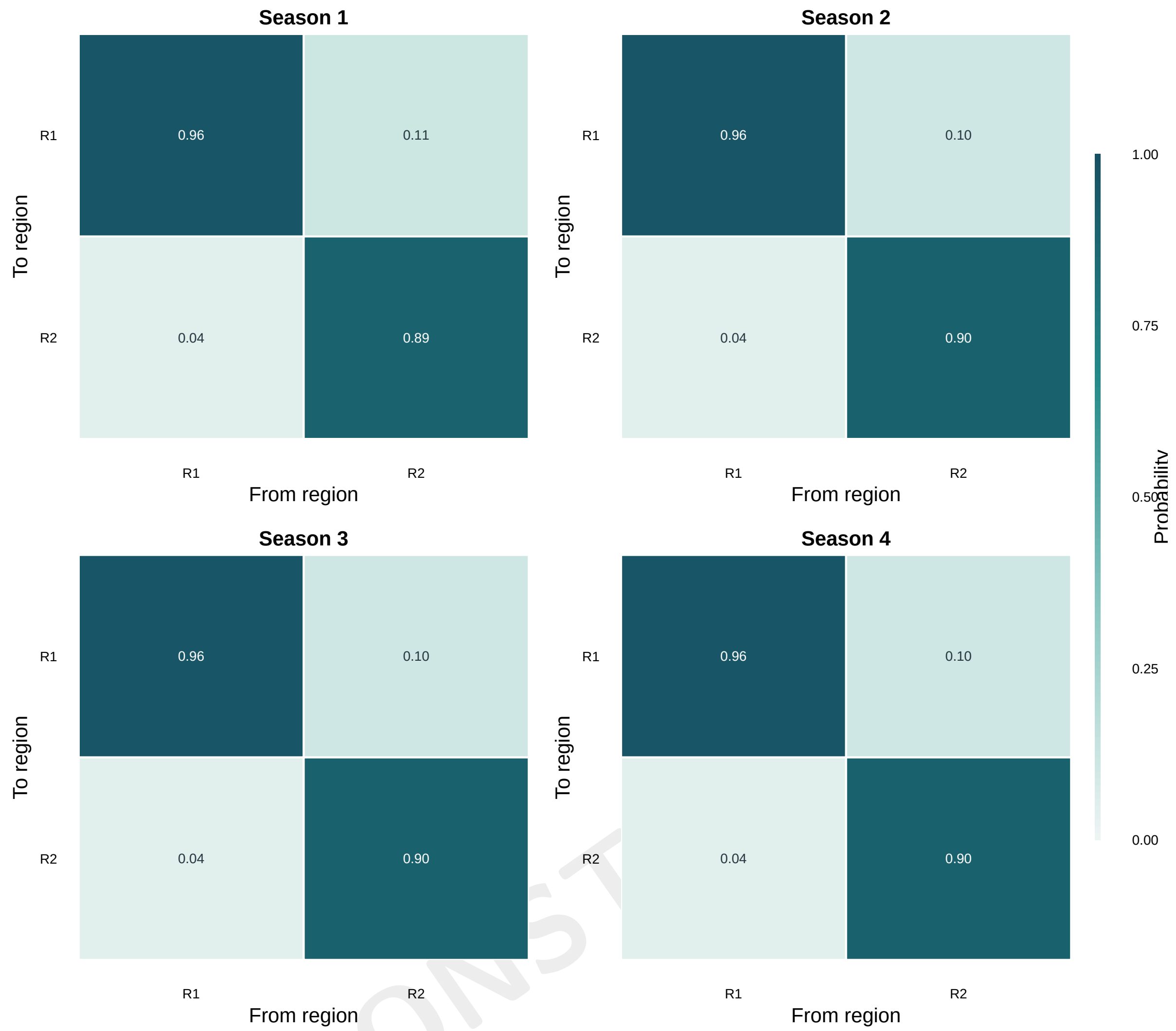

Figure 22: Mean movement probabilities across supplied model age classes, by season and source-to-destination region. Diagonal cells retain the probability of remaining in the source region.

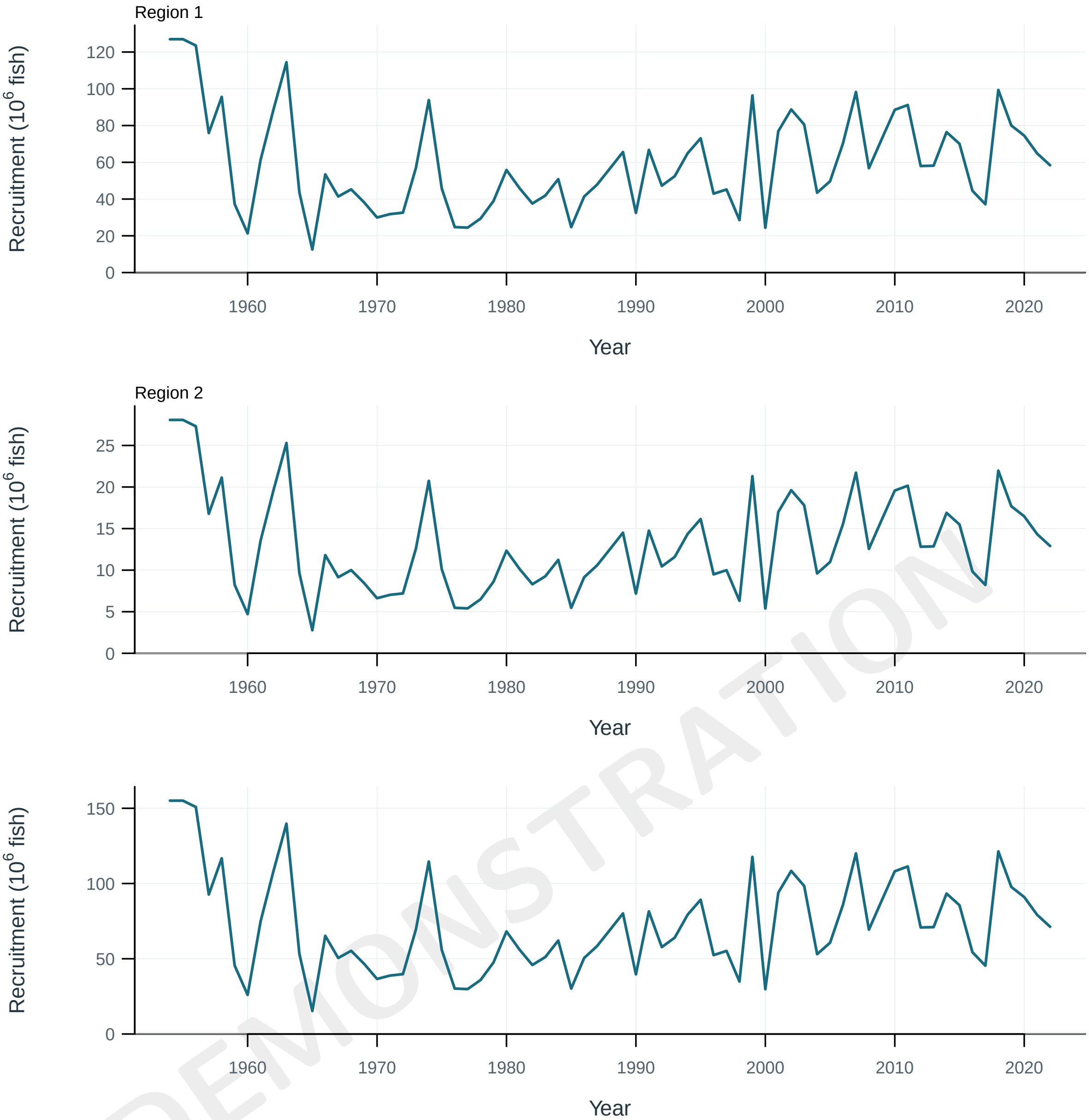

Figure 23: Annual recruitment overall and by model region.

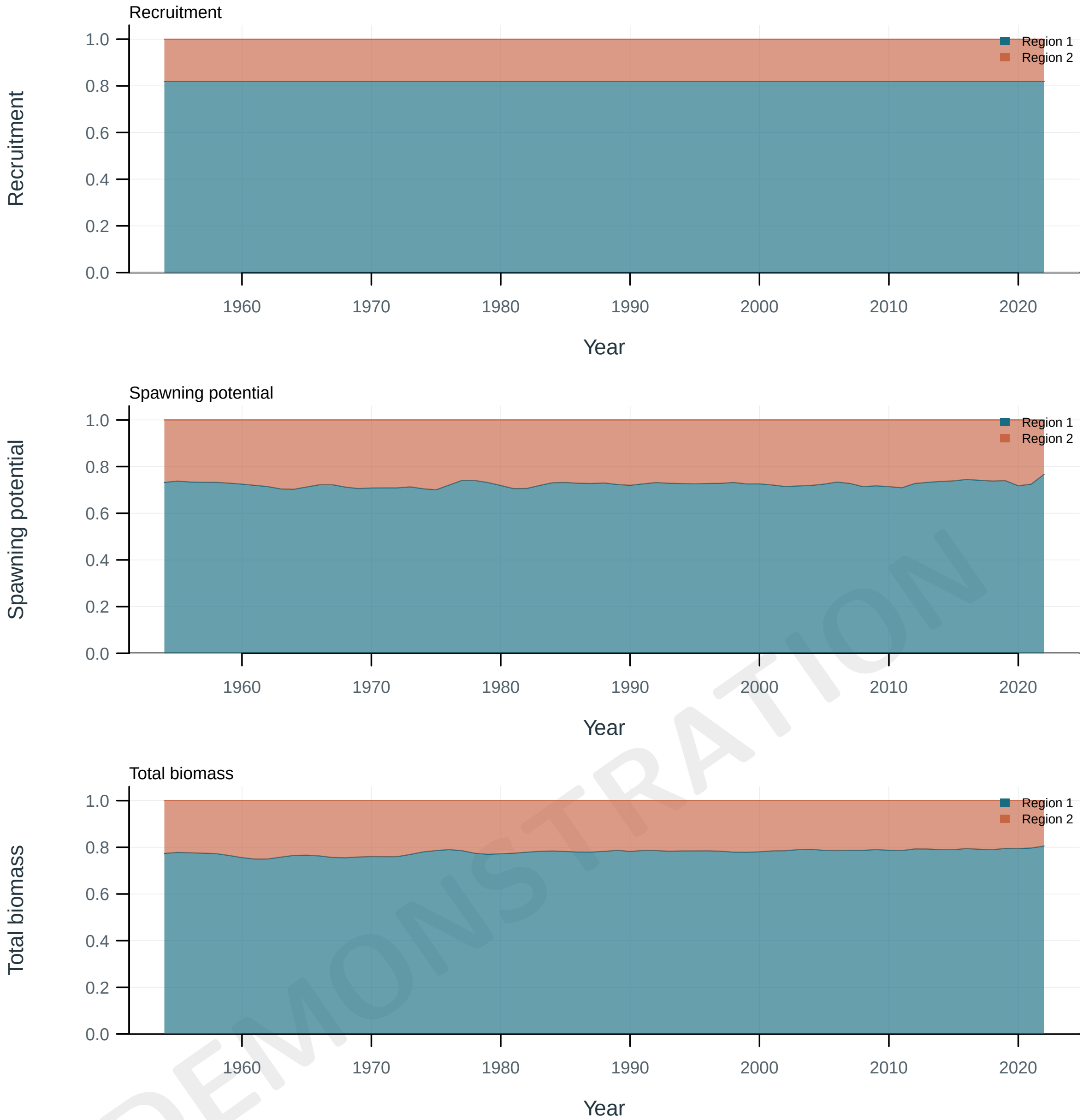

Figure 24: Regional shares of population quantities.

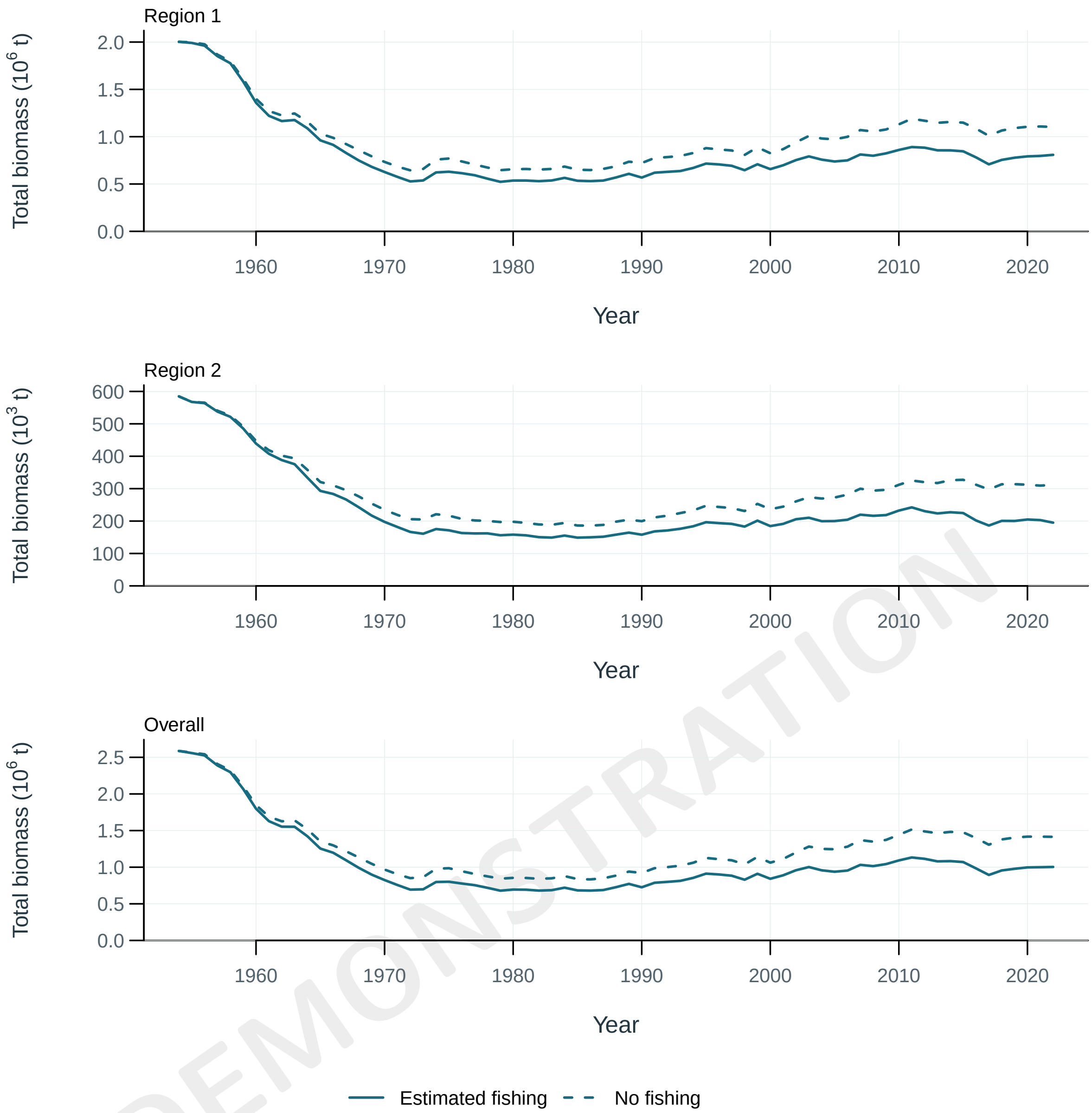

Figure 25: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

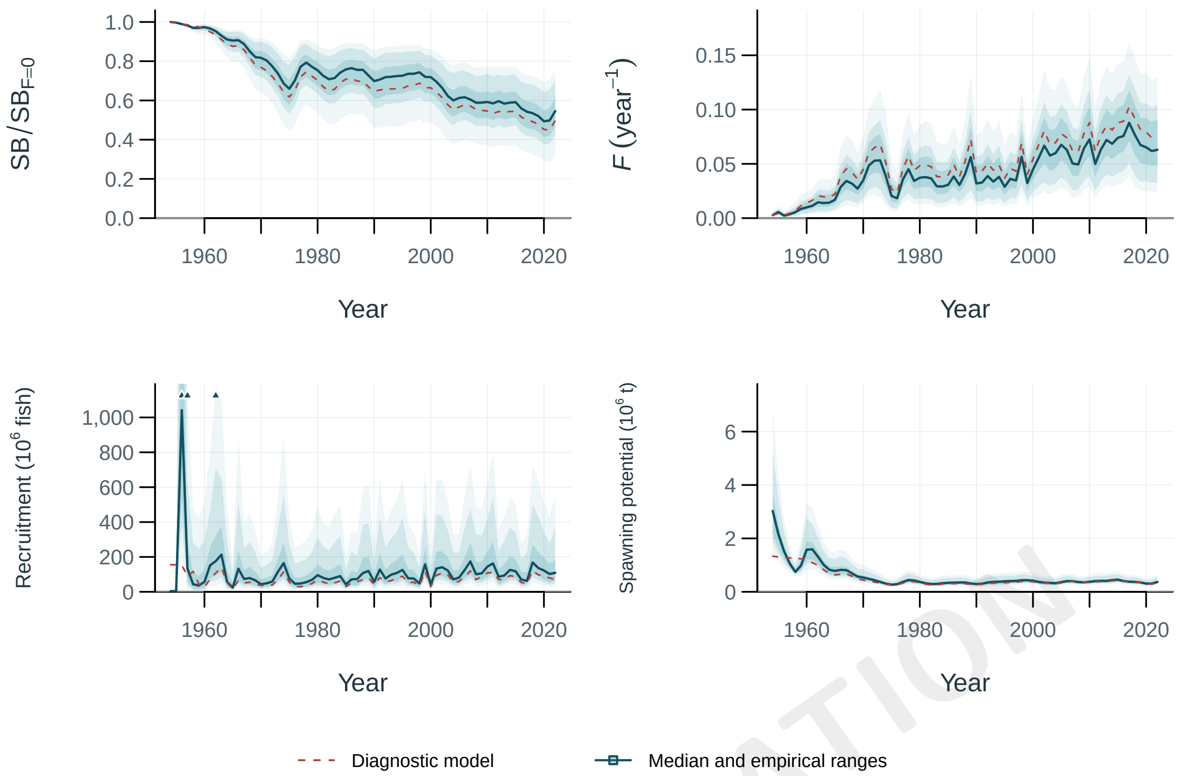

Figure 26: Empirical ensemble ranges and medians with the reference model (red dashed). Individual trajectories remain selectable in the supplementary viewer. Shading shows central 50, 80, 95% pointwise run ranges; the dark line is the median and the dashed red line is the reference model. Where isolated outer ranges dominate, upward triangles mark values beyond the robust static axis; complete values remain in the viewer.

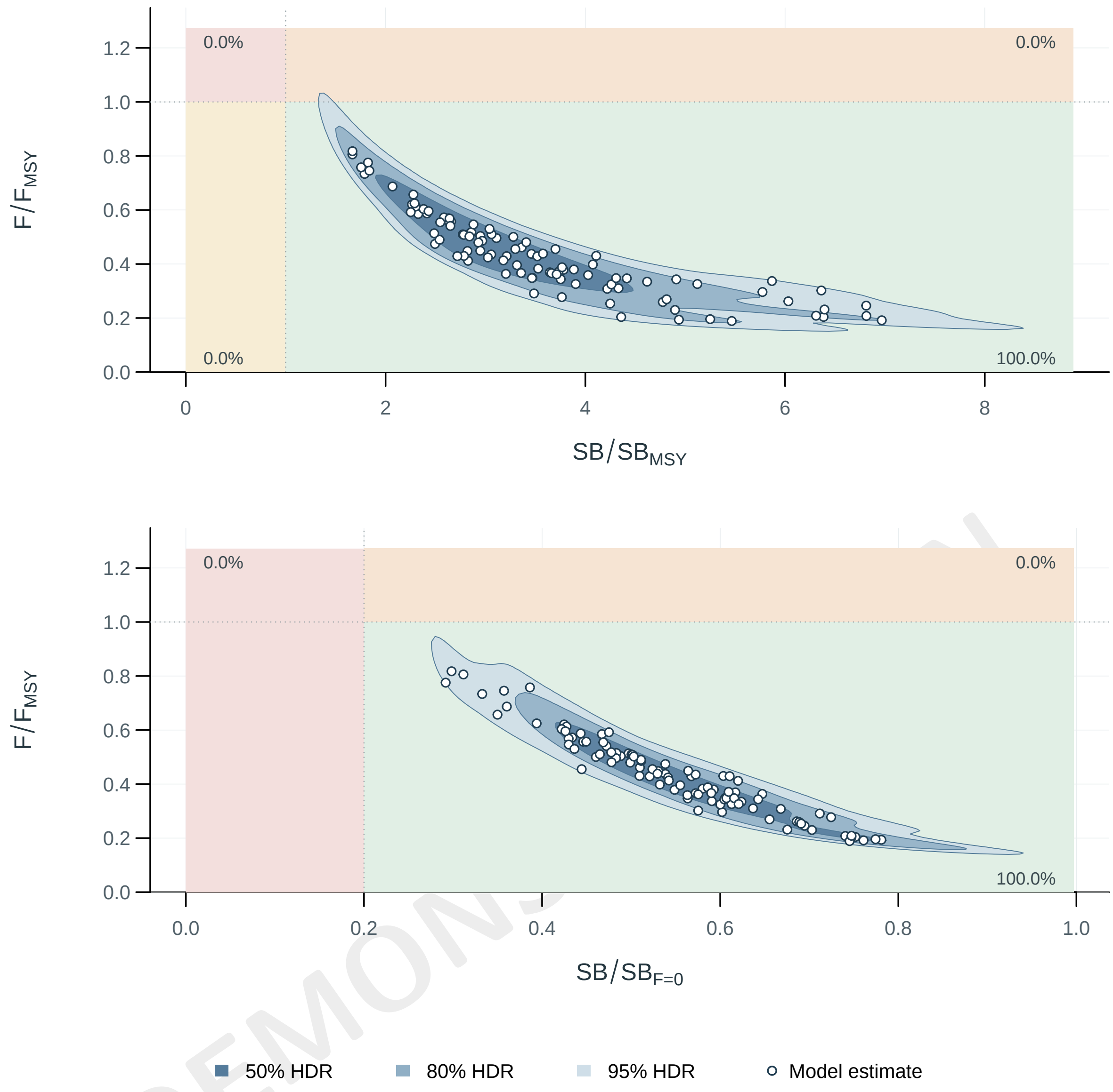

Figure 27: Kobe (upper) and Majuro (lower) status in 2022. Open points: model estimates; percentages: weighted status probabilities. Contours: 50%, 80% and 95% smoothed joint highest-density regions of structural model spread; smoothing can cross a management boundary even when its empirical percentage rounds to zero. Majuro probabilities distinguish depletion below the limit, and F above or below FMSY when the depletion criterion is met.

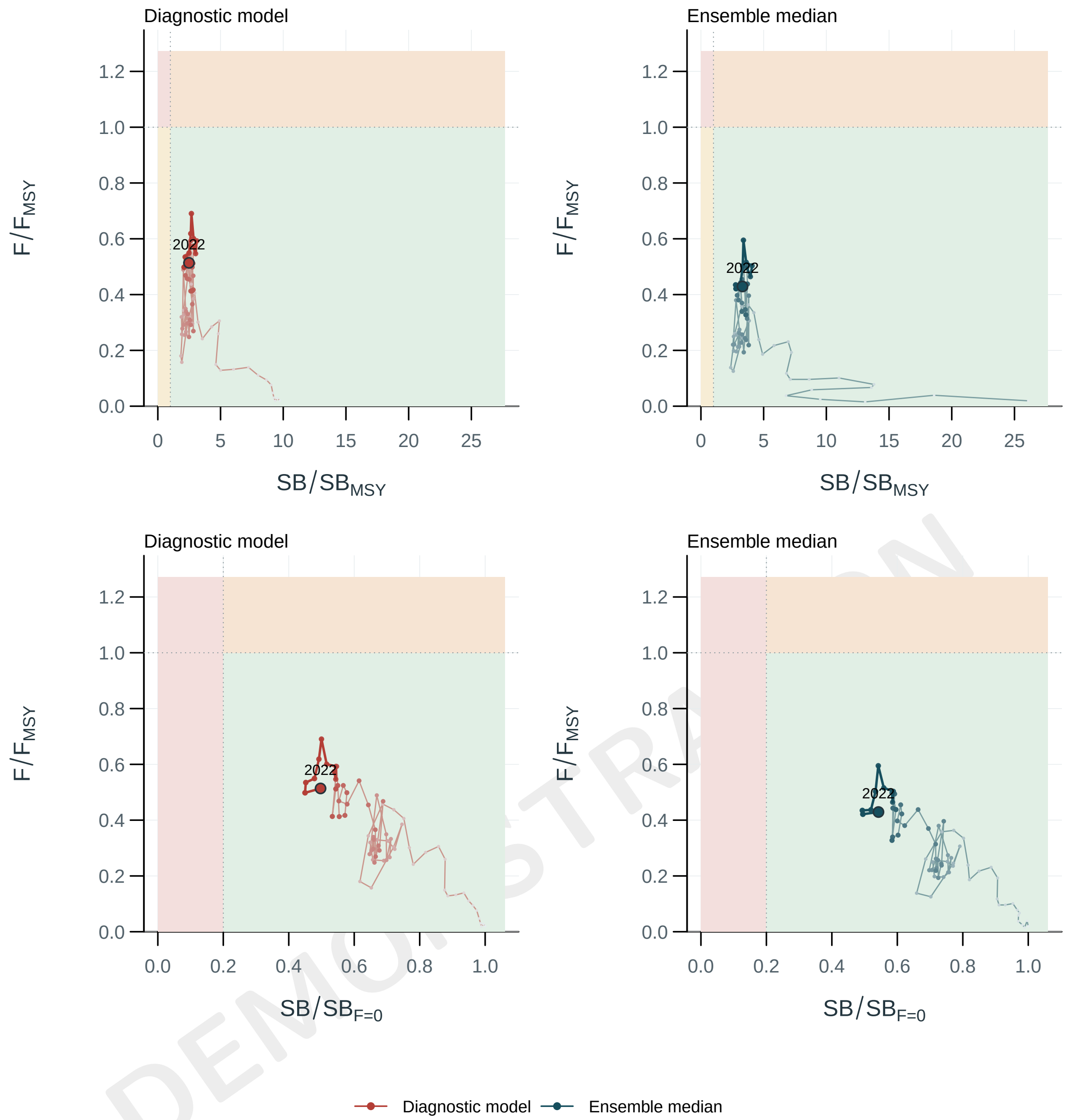

Figure 28: Stock status relative to the supplied spawning and fishing reference points. Rows show Kobe and Majuro coordinates; columns compare the diagnostic path with the annual ensemble median. Older years are muted, the latest ten years are emphasized and the terminal year is labelled. Individual ensemble-member paths remain selectable in the interactive viewer.

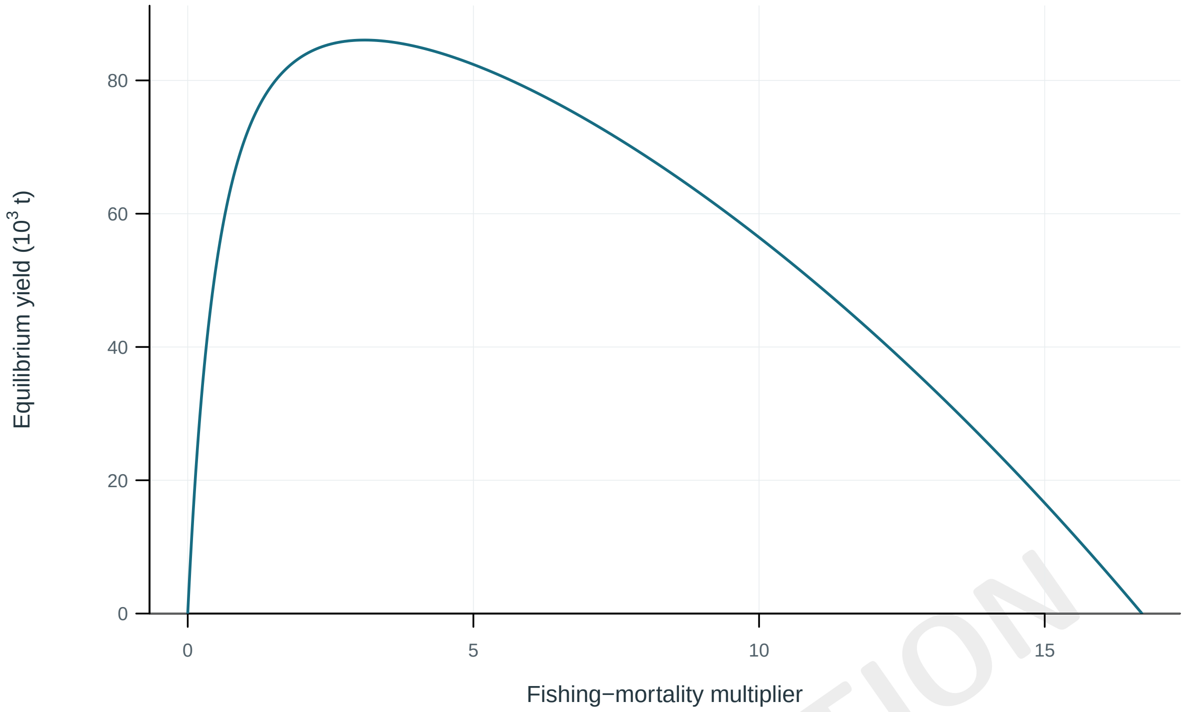

Figure 29: Equilibrium yield across fishing-mortality multipliers.

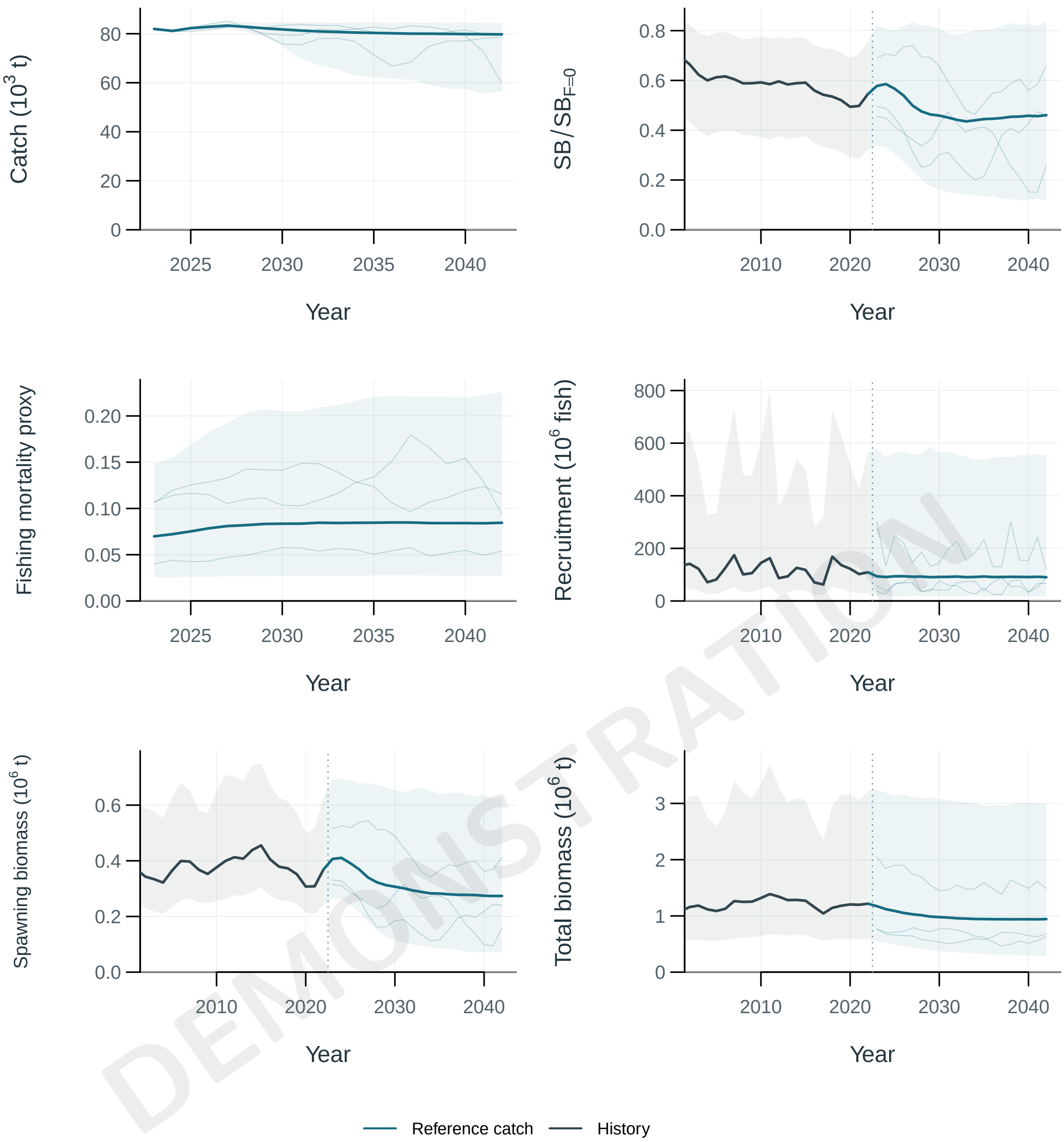

Figure 30: Historical and projected quantities by scenario with supplied uncertainty intervals. Central 95% intervals (2.5% to 97.5% quantiles across Monte Carlo ensemble members and recruitment replicates). Thin lines are a declared subset of individual projection paths. Static panels pair the full projection with a history window of equal duration; the viewer retains the complete history. Where isolated outer intervals dominate, upward triangles mark values beyond the robust static axis; complete values remain in the viewer.

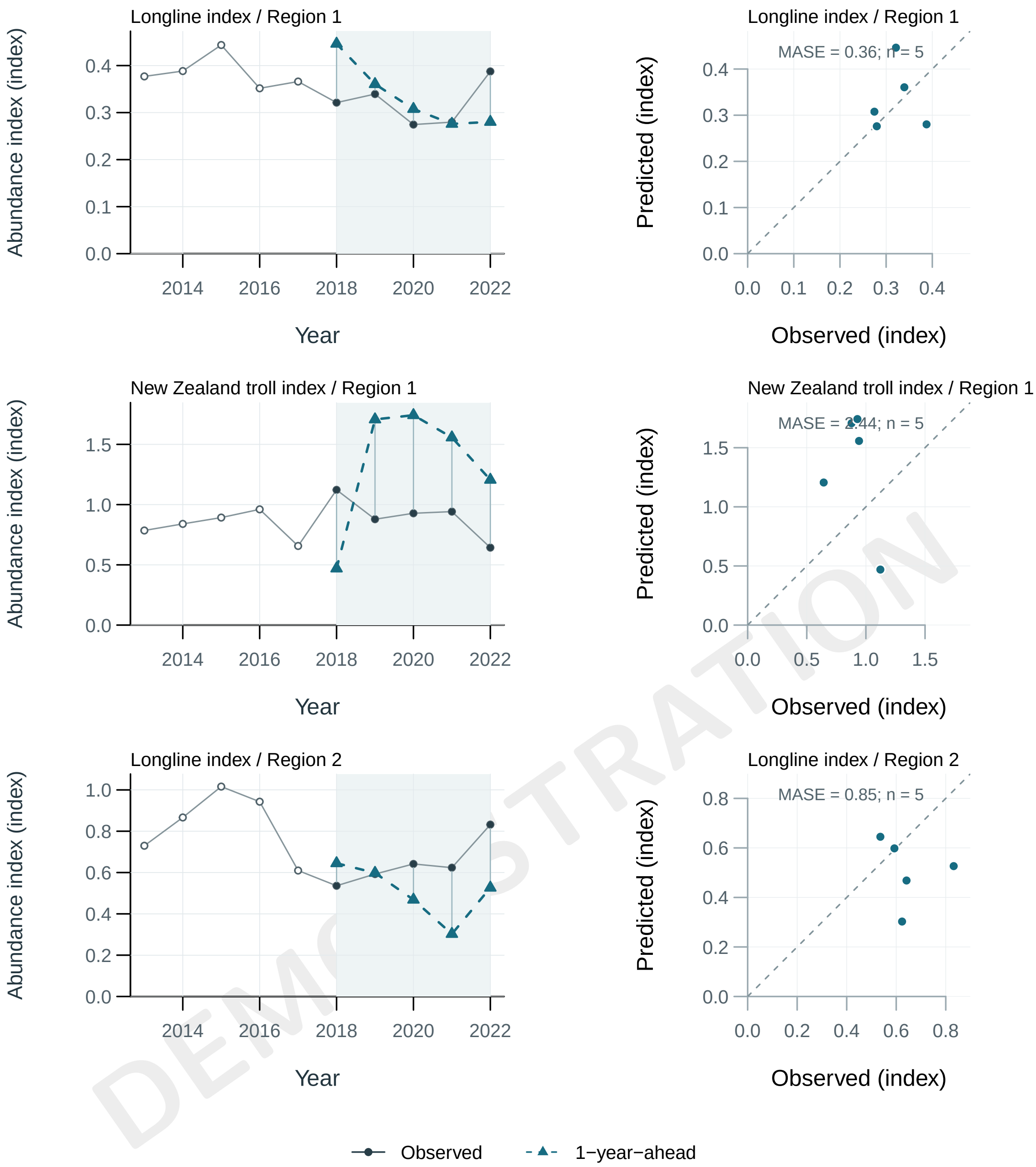

Figure 31: Observed values (black) and rolling 1-year-ahead predictions (teal); shading marks the held-out evaluation period. Right panels compare predictions with observations against the 1:1 line. MASE scales forecast errors by each origin's training naive MAE; n is the number of evaluated predictions.

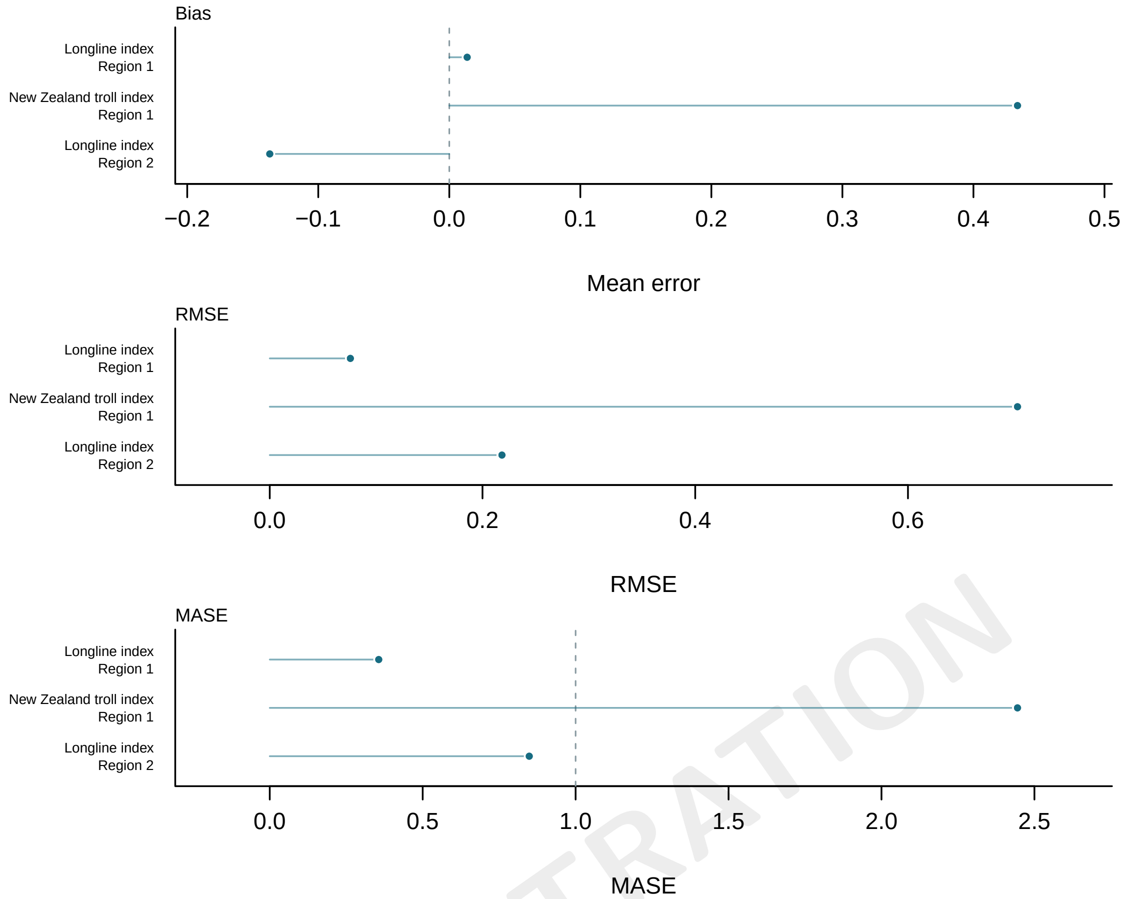

Figure 32: Out-of-sample hindcast error by index series and forecast lead. Points show bias and RMSE; MASE is scaled by the training-history naive mean absolute error and the dashed line marks one. The dashed bias reference marks zero. Values use only observations withheld at each rolling origin; n is retained in the accompanying table.

4 Supplementary outputs

Supplementary interactive viewer